

## On the malice of Herodotus



<sup>1</sup> The style, O Alexander, of Herodotus, as being simple, free, and easily suiting itself to its subject, has deceived many; but more, a persuasion of his dispositions being equally sincere. For it is not only (as Plato says) an extreme injustice, to make a show of being just when one is not so; but it is also the highest malignity, to pretend to simplicity and mildness and be in the meantime really most malicious. Now since he principally exerts his malice against the Boeotians and Corinthians, though without sparing any other, I think myself obliged to defend our ancestors and the truth against this part of his writings, since those who would detect all his other lies and fictions would have need of many books. But, as Sophocles has it, the face of persuasion, is prevalent, especially when delivered in the good language, and such as has power to conceal both the other absurdities and the ill-nature of the writer. King Philip told the Greeks who revolted from him to Titus Quinctius that they had got a more polished, but a longer lasting yoke. So the malice of Herodotus is indeed more polite and delicate than that of Theopompus, yet it pinches closer, and makes a more severe impression, — not unlike to those winds which, blowing secretly through narrow chinks, are sharper than those that are more diffused. Now it seems to me very convenient to delineate, as it were, in the rough draught, those signs and marks that distinguish a malicious narration from a candid and unbiassed one, applying afterwards every point we shall examine to such as appertain to them.

<sup>2</sup> First then, whoever in relating a story shall use the odious terms when gentler expressions might do as well, is it not to be esteemed impartial, but an enjoyer of his own fancy, in putting the worst construction on things; as if any one, instead of saying Nicias is too superstitious, should call him fanatic, or should accuse Cleon of presumption and madness rather than of inconsiderateness in speech.

<sup>3</sup> Secondly, when a writer, catching hold of a fault which has no reference to his story, shall draw it into the relation of such affairs as need it not, extending his narrative with circumlocutions, only that he may insert a man's misfortune, offence, or discommendable action, it is manifest that he delights in speaking evil. Therefore Thucydides would not clearly relate the faults of Cleon, which were very numerous; and as for Hyperbolus the orator, having touched at him in a word and called him an ill man, he let him go. Philistus also passed over all those outrages committed by Dionysius on the barbarians which had no connection with the Grecian affairs. For the excursions and digressions of history are principally allowed for fables and antiquities, and sometimes also for encomiums. But he who makes reproaches and detractions an addition to his discourse seems to incur the tragedian's curse on the "collector of men's calamities."

4 Now the opposite to this is known to every one, as the omitting to relate some good and laudable action, which, though it may seem not to be reprehensible, yet is then done maliciously when the omission happens in a place that is pertinent to the history. For to praise unwillingly is so far from being more civil than to dispraise willingly, that it is perhaps rather more uncivil.

5 The fourth sign of a partial disposition in writing of history I take to be this: When a matter is related in two or more several manners, and the historian shall embrace the worst. Sophisters indeed are permitted, for the obtaining either of profit or reputation, to undertake the defence of the worst cause; for they neither create any firm belief of the matter, nor yet do they deny that they are often pleased in maintaining paradoxes and making incredible things appear probable. But an historian is then just, when he asserts such things as he knows to be true, and of those that are uncertain reports rather the better than the worse. Nay, there are many writers who wholly omit the worse. Thus Ephorus writes of Themistocles, that he was acquainted with the treason of Pausanias and his negotiations with the King's lieutenants, but that he neither consented to it, nor hearkened to Pausanias's proffers of making him partaker of his hopes; and Thucydides left the whole matter out of his story, as judging it to be false.

6 Moreover, in things confessed to have been done, but for doing which the cause and intention is unknown, he who casts his conjectures on the worst side is partial and malicious. Thus do the comedians, who affirm the Peloponnesian war to have been kindled by Pericles for the love of Aspasia or the sake of Phidias, and not through any desire of honor, or ambition of pulling down the Peloponnesian pride and giving place in nothing to the Lacedaemonians. For those who suppose a bad cause for laudable works and commendable actions, endeavoring by calumnies to insinuate sinister suspicions of the actor when they cannot openly discommend the act, — as they that impute the killing of Alexander the tyrant by Theba not to any magnanimity or hatred of vice, but to a certain feminine jealousy and passion, and those that say Cato slew himself for fear Caesar should put him to a more shameful death, — such as these are manifestly in the highest degree envious and malicious.

7 An historical narration is also more or less guilty of malice, according as it relates the manner of the action; as if one should be said to have performed an exploit rather by money than bravery, as some affirm of Philip; or else easily and without any labor, as it is said of Alexander; or else not by prudence, but by Fortune, as the enemies of Timotheus painted cities falling into his nets as he lay sleeping. For they undoubtedly diminish the greatness and beauty of the actions, who deny the performer of them to have done them generously, industriously, virtuously, and by themselves.

8 Moreover, those who will directly speak ill of any one incur the reproach of moroseness, rashness, and madness, unless they keep within measure. But they who send forth calumnies obliquely, as if they were shooting arrows out of corners, and then stepping back think to conceal themselves by saying they do not believe what they most

earnestly desire to have believed, whilst they disclaim all malice, condemn themselves also of farther disingenuity.

<sup>9</sup> Next to these are they who with their reproaches intermix some praises, as did Aristoxenus, who, having termed Socrates unlearned, ignorant, and libidinous, added, Yet was he free from injustice. For, as they who flatter artificially and craftily sometimes mingle light reprehensions with their many and great praises, joining this liberty of speech as a sauce to their flattery; so malice, that it may gain belief to its accusations, adds also praise.

<sup>10</sup> We might here also reckon up more notes; but these are sufficient to let us understand the nature and manners of Herodotus.

<sup>11</sup> First therefore, — beginning, as the proverb is, with Vesta, — whereas all the Grecians affirm Io, daughter to Inachus, to have been worshipped with divine honor by the barbarians, and by her glory to have left her name to many seas and principal ports, and to have given a source and original to most noble and royal families; this famous author says of her, that she gave herself to certain Phoenician merchants, having been not unwillingly deflowered by a mariner, and fearing lest she should be found by her friends to be with child (Herodotus, i, 5). And he belies the Phoenicians as having delivered these things of her, and says that the Persian stories testify of her being carried away by the Phoenicians with other women (Herodotus, i, 1). Presently after, he gives sentence on the bravest and greatest exploits of Greece, saying that the Trojan war was foolishly undertaken for an ill woman. For it is manifest, says he, that had they not been willing they had never been ravished (Herodotus, i, 4). Let us then say, that the gods also acted foolishly, in inflicting their indignation on the Spartans for abusing the daughters of Scedasmus the Leuctrian, and in punishing Ajax for the violation of Cassandra. For it is manifest, if we believe Herodotus, that if they had not been willing they had never been defiled. And yet he himself said that Aristomenes was taken alive by the Spartans; and the same afterwards happened to Philopoemen, general of the Achaeans; and the Carthaginians took Regulus, the consul of the Romans; than whom there are not easily to be found more valiant and warlike men. Nor is it to be wondered, since even leopards and tigers are taken alive by men. But Herodotus blames the poor women that have been abused by violence, and patronizes their ravishers.

<sup>12</sup> Nay, he is so favorable to the barbarians, that, acquitting Busiris of those human sacrifices and that slaughter of his guests for which he is accused, and attributing by his testimony to the Egyptians much religion and justice, he endeavors to cast that abominable wickedness and those impious murders on the Grecians. For in his Second Book he says, that Menelaus, having received Helen from Proteus and having been honored by him with many presents, showed himself a most unjust and wicked man; for wanting a favorable wind to set sail, he found out an impious device, and having taken two of the inhabitants' boys, consulted their entrails; for which villany being hated and persecuted, he fled with his ships directly into Libya (Herodotus, ii, 45). From what

Egyptian this story proceeds, I know not. For, on the contrary, many honors are even at this day given by the Egyptians to Helen and Menelaus.

<sup>13</sup> The same Herodotus, that he may still be like himself, says that the Persians learned the defiling of the male sex from the Greeks (Herodotus, i, 135). And yet how could the Greeks have taught this impurity to the Persians, amongst whom, as is confessed by many, boys had been castrated before ever they arrived in the Grecian seas? He writes also, that the Greeks were instructed by the Egyptians in their pomps, solemn festivals, and worship of the twelve gods; that Melampus also learned of the Egyptians the name of Dionysus (or Bacchus) and taught it the other Greeks; that the mysteries likewise and rites of Ceres were brought out of Egypt by the daughters of Danaus; and that the Egyptians were wont to beat themselves and make great lamentation, but yet he himself refused to tell the names of their deities, but concealed them in silence. As to Hercules and Bacchus, whom the Egyptians named gods, and the Greeks very aged men, he nowhere has such scruples and hesitation; although he places also the Egyptian Hercules amongst the gods of the second rank, and Bacchus amongst those of the third, as having had some beginning of their being and not being eternal, and yet he pronounces those to be gods; but to the gods Bacchus and Hercules, as having been mortal and being now demi-gods, he thinks we ought to perform anniversary solemnities, but not to sacrifice to them as to gods. The same also he said of Pan, overthrowing the most venerable and purest sacrifices of the Greeks by the proud vanities and mythologies of the Egyptians. (For the passages referred to in this chapter, see Herodotus, ii, 48, 51, 145, 146, 171.)

<sup>14</sup> Nor is this impious enough; but moreover, deriving the pedigree of Hercules from Perseus, he says that Perseus was an Assyrian, as the Persians affirm. "But the leaders," says he, "of the Dorians may appear to be descended in a right line from the Egyptians, reckoning their ancestors from before Danae and Acrisius." (Herodotus, vi. 53, 54). Here he has wholly passed by Epaphus, Io, Iasus, and Argus, being ambitious not only to make the other Herculese Egyptians and Phoenicians but to carry this also, whom himself declares to have been the third, out of Greece to the barbarians. But of the ancient learned writers, neither Homer, nor Hesiod, nor Archilochus, nor Pisander, nor Stesichorus, nor Alcman, nor Pindar, makes any mention of the Egyptian or the Phoenician Hercules, but all acknowledge this our own Boeotian and Argive Hercules.

<sup>15</sup> Now of the seven sages, whom he calls Sophisters, he affirms Thales to have been a barbarian, descended of the Phoenicians (Herodotus, i, 170). Speaking ill also of the gods under the person of Solon, he has these words: "Thou, O Croesus, askest me concerning human affairs, who know that every one of the deities envious and tumultuous." (Herodotus, i, 32) Thus attributing to Solon what himself thinks of the gods, he joins malice to blasphemy.

Having made use also of Pittacus in some trivial matters, not worth the mentioning, he has passed over the greatest and gallantest action that was ever done by him. For when the Athenians and Mitylenaeans were at war about Sigaeum, Phrynon, the

Athenian general, challenging whoever would come forth to a single combat, Pittacus advanced to meet him, and catching him in a net, slew that stout and giant-like man; for which when the Mitylaenans offered him great presents, darting his javelin as far as he could out of his hand, he desired only so much ground as he should reach with that throw; and the place is to this day called Pittacium. Now what does Herodotus, when he comes to this? Instead of Pittacus's valiant act, he tells us the fight of Alcaeus the poet, who throwing away his arms ran out of the battle; by thus not writing of honorable deeds and not passing over such as are dishonorable, he offers his testimony to those who say, that from one and the same malice proceed both envy and a rejoicing at other men's harms (Herodotus v. 95) .

<sup>16</sup> After this, he accuses of treason the Alcmaeonidae who showed themselves generous men, and delivered their country from tyranny (Herodotus, i, 61). He says, that they received Pisistratus after his banishment and got him called home, on condition he should marry the daughter of Megacles; but the damsel saying to her mother, Do you see, mother, how I am known by Pisistratus contrary to nature? The Alcmaeonidae were so offended at this villany, that they expelled the tyrant.

<sup>17</sup> Now that the Lacedaemonians might have no less share of his malice than the Athenians, behold how he bespatters Othryadas, the man most admired and honored by them. "He only," says Herodotus, "remaining alive of the three hundred, and ashamed to return to Sparta, his companions being lost, slew himself on the spot at Thyrae." (Herodotus, i, 82). For having before said the victory was doubtful on both sides, he here, by making Othryadas ashamed, witnesses that the Lacedaemonians were vanquished. For it was shameful for him to survive, if conquered; but glorious, if conqueror.

<sup>18</sup> I pass by now, that having, represented Croesus as foolish, vainglorious, and ridiculous in all things, he makes him, when a prisoner, to have taught and instructed Cyrus, who seems to have excelled all other kings in prudence, virtue, and magnanimity (Herodotus, i, 155, 156, 207, and 208). Having testified of the same Croesus nothing else that was commendable but his honoring the gods with many and great oblations, he shows that very act of his to have been the most impious of all. For he says, that he and his brother Pantoleon contended for the kingdom while their father was yet alive; and that Croesus, having obtained the crown, caused a companion and familiar friend of Pantoleon's to be torn in pieces in a fulling-mill, and sent presents to the gods from his property (Herodotus, i, 92). Of Deioeces also, the Median, who by virtue and justice obtained the government, he says that he got it not by real but pretended justice (Herodotus, i, 96).

<sup>19</sup> But I let pass the barbarian examples, since he has offered us plenty enough in the Grecian affairs. He says, that the Athenians and many other Ionians were so ashamed of that name that they wholly refused to be called Ionians; and that those who esteemed themselves the noblest among them, and who had come forth from the very Prytaneum of Athens, begat children on barbarian wives whose parents, husbands, and former

children they had slain; that the women had therefore made a law among themselves, confirmed it by oath, and delivered it to be kept by their daughters, never to eat with their husbands, nor to call any of them by his name; and that the present Milesians are descended from these women. Having afterwards added that those are true Ionians who celebrate the feast called Apaturia; they all, says he, keep it except the Ephesians and Colophonians (Herodotus, i, 143–148). In this manner does he deprive these two states of their nobility.

20 He says moreover, that the Cumaeans and Mitylenaeans agreed with Cyrus to deliver up to him for a price Pactyas, who had revolted from him. I know not indeed, says he, for how much; since it is not certain what it was. Bravo! — not to know what it was, and yet to cast such an infamy on a Grecian city, without an assured knowledge! He says farther, that the Chians took Pactyas, who was brought to them out of the temple of Minerva Poliuchus (or Guardianess of the city), and delivered him up, having received the city Atarneus for their recompense. And yet Charon the Lampsacenan, a more ancient writer, relating this matter concerning Pactyas, charges neither the Mitylenaeans nor the Chians with any such action. These are his very words: “Pactyas, hearing that the Persian army drew near, fled first to Mitylene, then to Chios, and there fell into the hands of Cyrus.” (See Herodotus, i, 157 etc.)

21 Our author in his Third Book, relating the expedition of the Lacedaemonians against the tyrant Polycrates, affirms, that the Samians think and say that the Spartans, to recompense them for their former assistance against the Messenians, both brought back the Samians that were banished, and made war on the tyrant; but that the Lacedaemonians deny this, and say, they undertook this design not to help or deliver the Samians, but to punish them for having taken away a cup sent by them to Croesus, and besides, a breastplate sent them by Amasis (Herodotus, iii, 47, 48). And yet we know that there was not at that time any city so desirous of honor, or such an enemy to tyrants, as Sparta. For what breastplate or cup was the cause of their driving the Cypselidae out of Corinth and Ambracia, Lygdamis out of Naxos, the children of Pisistratus out of Athens, Aeschines out of Sicyon, Symmachus out of Thasus, Aulis out of Phocis, and Aristogenes out of Miletus; and of their overturning the domineering powers of Thessaly, pulling down Aristomedes and Angelus by the help of King Leotychides? — which facts are elsewhere more largely described. Now, if Herodotus says true, they were in the highest degree guilty both of malice and folly, when, denying a most honorable and most just cause of their expedition, they confessed that in remembrance of a former injury, and too highly valuing an inconsiderable matter, they invaded a miserable and afflicted people.

22 Now perhaps he gave the Lacedaemonians this stroke, as directly falling under his pen; but the city of Corinth, which was wholly out of the course of his story, he has brought in-going out of his way (as they say) to fasten upon it — and has bespattered it with a most filthy crime and most shameful calumny. “The Corinthians,” says he, “studiously helped this expedition of the Lacedaemonians to Samos, as having

themselves also been formerly affronted by the Samians.” The matter was this. Periander tyrant of Corinth sent three hundred boys, sons to the principal men of Corcyra, to King Alyattes, to be gelt. These, going ashore in the island of Samos, were by the Samians taught to sit as suppliants in the temple of Diana, where they preserved them, setting before them for their food sesame mingled with honey. This our author calls an affront put by the Samians on the Corinthians, who therefore instigated the Lacedaemonians against them, to wit, because the Samians had saved three hundred children of the Greeks from being unmanned. By attributing this villany to the Corinthians, he makes the city more wicked than the tyrant. He indeed was revenging himself on those of Corcyra who had slain his son; but what had the Corinthians suffered, that they should punish the Samians for putting an obstacle to so great a cruelty and wickedness? — and this, after three generations, reviving the memory of an old quarrel for the sake of that tyranny, which they found so grievous and intolerable that they are still endlessly abolishing all the monuments and marks of it, though long since extinct. Such then was the injury done by the Samians to the Corinthians. Now what a kind of punishment was it the Corinthians would have inflicted on them? Had they been indeed angry with the Samians, they should not have incited the Lacedaemonians, but rather diverted them from their war against Polycrates, that the Samians might not by the tyrant’s overthrow recover liberty, and be freed from their slavery. But (what is most to be observed) why were the Corinthians so offended with the Samians, that desired indeed but were not able to save the Corcyraeans children, and yet were not displeased with the Cnidians, who both preserved them and restored them to their friends? Nor indeed have the Corcyraeans any great esteem for the Samians on this account; but of the Cnidians they preserve a grateful recollection, having granted them several honors and privileges, and made decrees in their favor. For these, sailing to Samos, drove away Periander’s guards from the temple, and taking the children aboard their ships, carried them safe to Corcyra; as it is recorded by Antenor the Cretan, and by Dionysius the Chalcidian in his foundations. Now that the Spartans undertook not this war on any design of punishing the Samians, but to save them by delivering them from the tyrant, we have the testimony of the Samians themselves. For they affirm that there is in Samos a monument erected at the public charge, and honors there done to Archias a Spartan, who fell fighting valiantly in that quarrel; for which cause also his posterity still keep a familiar and friendly correspondence with the Samians, as Herodotus himself witnesses. (See Herodotus, iii. 48.)

<sup>23</sup> In his Fifth Book, he says, that Clisthenes, one of the best and noblest men in Athens, persuaded the priestess Pythia to be a false prophetess, and always to exhort the Lacedaemonians to free Athens from the tyrants; calumniating this most excellent and just action by the imputation of so great a wickedness and imposture, and taking from Apollo the credit of that true and good prophecy, beseeeming even Themis herself, who is also said to have joined with him. He says farther, that Isagoras prostituted his wife to Cleomenes, who came to her (Herodotus, v. 63, 70). Then, as his manner is, to gain

credit by mixing some praises with his reproaches, he says: Isagoras the son of Tisander was of a noble family, but I cannot tell the original of it; his kinsmen, however, sacrifice to the Carian Jupiter (Herodotus, v. 66). O this pleasant and cunning scoffer of a writer, who thus disgracefully sends Isagoras to the Carians, as it were to the ravens. As for Aristogiton, he puts him not forth at the back door, but thrusts him directly out of the gate into Phoenicia, saying that he had his original from the Gephyraeans, and that the Gephyraeans were not, as some think, Euboeans or Eretrians, but Phoenicians, as himself has heard by report (Herodotus, v. 58). And since he cannot altogether take from the Lacedaemonians the glory of having delivered the Athenians from the tyrants, he endeavors to cloud and disgrace that most honorable act by as foul a passion. For he says, they presently repented of it, as not having done well, in that they had been persuaded by spurious and deceitful oracles to drive the tyrants, who were their allies and had promised to put Athens into their hands, out of their country, and had restored the city to an ungrateful people. He adds, that they were about to send for Hippias from Sigeum, and bring him back to Athens; but that they were opposed by the Corinthians, Sosicles telling them how much the city of Corinth had suffered under the tyranny of Cypselus and Periander (Herodotus, v. 90, 91). And yet there was no outrage of Periander's more abominable and cruel than his sending the three hundred children to be emasculated, for the delivering and saying of whom from that contumely the Corinthians, he says, were angry and bore a grudge against the Samians, as having put an affront upon them. With so much repugnance and contradiction is that malice of his discourse filled, which on every occasion insinuates itself into his narrations.

<sup>24</sup> After this, relating the action of Sardis, he, as much as in him lies, diminishes and discredits the matter; being so audacious as to call the ships which the Athenians sent to the assistance of the Ionians, who had revolted from the King the beginning of evils, because they endeavored to deliver so many and so great Grecian cities from the barbarians (Herodotus, v. 97). As to the Eretrians, making mention of them only by the way, he passes over in silence a great, gallant, and memorable action of theirs. For when all Ionia was in a confusion and uproar, and the King's fleet drew nigh, they, going forth to meet him, overcame in a sea-fight the Cyprians in the Pamphylian Sea. Then turning back and leaving their ships at Ephesus, they invaded Sardis and besieged Artaphernes, who was fled into the castle, that so they might raise the siege of Miletus. And this indeed they effected, causing the enemies to break up their camp and remove thence in a wonderful fright, and then seeing themselves in danger to be oppressed by a multitude, retired. This not only others, but Lysanias of Mallus also in his history of Eretria relates, thinking it convenient, if for no other reason, yet after the taking and destruction of the city, to add this valiant and heroic act. But this writer of ours says, they were defeated, and pursued even to their ships by the barbarians; though Charon the Lampsacenean has no such thing, but writes thus, word for word: "The Athenians set forth with twenty galleys to the assistance of the Ionians, and going to Sardis, took all thereabouts, except the King's wall; which having done, they returned to Miletus."



<sup>25</sup> In his Sixth Book, our author, discoursing of the Plataeans, — how they gave themselves to the Lacedaemonians, who exhorted them rather to have recourse to the Athenians, who were nearer to them and no bad defenders, — adds, not as a matter of suspicion or opinion, but as a thing certainly known by him, that the Lacedaemonians gave the Plataeans this advice, not so much for any goodwill, as through a desire to find work for the Athenians by engaging them with the Boeotians (Herodotus, vi. 108). If then Herodotus is not malicious, the Lacedaemonians must have been both fraudulent and spiteful; and the Athenians fools, in suffering themselves to be thus imposed on; and the Plataeans were brought into play, not for any good-will or respect, but as an occasion of war.

<sup>26</sup> He is farther manifestly convinced of belying the Lacedaemonians, when he says that, whilst they expected the full moon, they failed of giving their assistance to the Athenians at Marathon. For they not only made a thousand other excursions and fights at the beginning of the month, without staying for the full moon; but wanted so little of being present at this very battle, which was fought the sixth day of the month Boedromion, that at their coming they found the dead still lying in the field. And yet he has written thus of the full moon: “It was impossible for them to do these things at that present, being unwilling to break the law; for it was the ninth of the month, and they said, they could not go forth on the ninth day, the orb of the moon being not yet full. And therefore they stayed for the full moon” (Herodotus, vi. 106). But thou, O Herodotus, transferest the full moon from the middle to the beginning of the month, and at the same time confoundest the heavens, days, and all things; and yet thou dost claim to be the historian of Greece!

And professing to write more particularly and carefully of the affairs of Athens, thou dost not so much as say a word of that solemn procession which the Athenians even at this day send to Agrae, celebrating a feast of thanksgiving to Hecate for their victory. But this helps Herodotus to refel the crime with which he is charged, of having flattered the Athenians for a great sum of money he received of them. For if he had rehearsed these things to them, they would not have omitted or neglected to remark that Philippides, when on the ninth he summoned the Lacedaemonians to the fight, must have come from it himself, since (as Herodotus says) he went in two days from Athens to Sparta; unless the Athenians sent for their allies to the fight after their enemies were overcome. Indeed Diyllus the Athenian, none of the most contemptible as an historian, says, that he received from Athens a present of ten talents, Anytus proposing the decree. Moreover Herodotus, as many say, has in relating the fight at Marathon derogated from the credit of it, by the number he sets down of the slain. For it is said that the Athenians made a vow to sacrifice so many kids to Diana Agrotera, as they should kill barbarians; but that after the fight, the number of the dead appearing infinite, they appeased the goddess by making a decree to immolate five hundred to her every year.

<sup>27</sup> But letting this pass, let us see what was done after the fight. “The barbarians,” say he, “retiring back with the rest of their ships, and taking the Eretrian slaves out of the

island, where they had left them, doubled the point of Sunium, desiring to prevent the Athenians before they could gain the city. The Athenians suspected this to have been done by a plot of the Alcmaeonidae, who by agreement showed a shield to the Persians when they were got into their ships. They therefore doubled the cape of Sunium.” (Herodotus, vi. 115, 121–124) Let us in this place take no notice of his calling the Eretrians slaves, who showed as much courage and gallantry in this war as any other of the Grecians, and suffered things unworthy their virtue. Nor let us insist much on the calumny with which he defames the Alcmaeonidae, some of whom were both the greatest families and noblest men of the city. But the greatness of the victory itself is overthrown, and the end of that so celebrated action comes to nothing, nor does it seem to have been a fight or any great exploit, but only a light skirmish with the barbarians, as the envious and ill-willers affirm, if they did not after the battle fly away, cutting their cables and giving themselves to the wind, to carry them as far as might be from the Attic coast, but having a shield lifted up to them as a signal of treason, made straight with their fleet for Athens, in hope to surprise it, and having at leisure doubled the point of Sunium, were discovered above the port Phalerum, so that the chief and most illustrious men, despairing to save the city would have betrayed it. For a little after, acquitting the Alcmaeonidae, he charges others with the treason. “For the shield indeed was shown, nor can it be denied,” says he, as if he had seen it himself. But this could no way be, since the Athenians obtained a solid victory; and if it had been done, it could not have been seen by the barbarians, flying in a hurry amidst wounds and arrows into their ships, and leaving every one the place with all possible speed. But when he again pretends to excuse the Alcmaeonidae of those crimes which he first of all men objected against them, and speaks thus: “I cannot credit the report that the Alcmaeonidae by agreement would ever have lifted up a shield to the Persians, and have brought the Athenians under the power of the barbarians and Hippias”; it reminds me of a certain proverbial saying, — Stay and be caught, crab, and I’ll let you go. For why art thou so eager to catch him, if thou wilt let him go when he is caught? Thus you first accuse, then apologize; and you write calumnies against illustrious men, which again you refute. And you discredit yourself; for you heard no one else but yourself say that the Alcmaeonidae lifted up a shield to the vanquished and flying barbarians. And in those very things which you allege for the Alcmaeonidae, you show yourself a sycophant. For if, as here you write, the Alcmaeonidae were more or no less enemies to tyrants than Callias, the son of Phaenippus and father of Hipponicus, where will you place their conspiracy, of which you write in your First Book, that assisting Pisistratus they brought him back from exile to the tyranny and did not drive him away till he was accused of unnaturally abusing his wife? Such then are the repugnances of these things; and by his intermixing the praises of Callias, the son of Phaenippus, amidst the crimes and suspicions of the Alcmaeonidae, and joining to him his son Hipponicus, who was (as Herodotus himself says) one of the richest men in Athens, he confesses that he brought in Callias not for any necessity of the story, but to ingratiate himself and gain favor with Hipponicus.

28 Now, whereas all know that the Argives denied not to enter into the common league of the Grecians, though they thought not fit to follow and be under the command of the Lacedaemonians, who were their mortal enemies, and that this was no otherways, our author subjoins a most malicious cause for it, writing thus: "When they saw they were comprised by the Greeks, knowing that the Lacedaemonians would not admit them into a share of the command, they requested it, that they might have a pretence to lie still." "And of this," he says, "the Argive ambassadors afterwards put Artaxerxes in mind, when they attended him at Susa, and the King said, he esteemed no city more his friend than Argos." Then adding, as his manner is, to cover the matter, he says: "Of these things I know nothing certainly; but this I know, that all men have faults, and that the worst things were not done by the Argives; but I must tell such things as are reported, though I am not bound to believe them all; and let this be understood of all my narrations. For it is farther said that the Argives, when they were not able to sustain the war against the Lacedaemonians, called the Persians into Greece, willing to suffer anything rather than the present trouble." (Herodotus, vii. 148–152) Therefore, as himself reports the Ethiopian to have said of the ointment and purple, "Deceitful are the beauties, deceitful the garments of the Persians," (Herodotus, iii, 22.) may not any one say also of him, Deceitful are the phrases, deceitful the figures of Herodotus's speeches; as being perplexed, unsound, and full of ambiguities? For as painters set off and render more eminent the luminous part of their pictures by adding shadows, so he by his denials extends his calumnies, and by his dubious speeches makes his suspicions take deeper impression. If the Argives joined not with the other Greeks, but stood out through an emulation of the Lacedaemonians command and valor, it cannot be denied but that they acted in a manner not beseeming their nobility and descent from Hercules. For it had been more honorable for the Argives under the leadership of Siphnians and Cythnians to have defended the Grecian liberty, than contending with the Spartans for superiority to have avoided so many and such signal combats. And if it was they who brought the Persians into Greece, because their war against the Lacedaemonians succeeded ill, how came it to pass, that they did not at the coming of Xerxes openly join themselves to the Medes? Or if they would not fight under the King, why did they not, being left at home, make incursions into Laconia or again attempt Thyreae or by some other way disturb and infest the Lacedaemonians? For they might have greatly damaged the Grecians, by hindering the Spartans from going with so great an army to Plataea.

29 But in this place indeed he has highly magnified the Athenians and pronounced them the saviours of Greece, doing herein rightly and justly, if he had not intermixed many reproaches with their praises. But now, when he says (Herodotus, vii. 139) that (but for the Athenians) the Lacedaemonians would have been betrayed by the other Greeks, and then, being left alone and having performed great exploits, they would have died generously; or else, having before seen that the Greeks were favoring the Medes, they would have made terms with Xerxes; it is manifest, he speaks not these things to the commendation of the Athenians, but he praises the Athenians that he may speak ill of all

the rest. For how can any one now be angry with him for so bitterly and intemperately upbraiding the Thebans and Phocians at every turn, when he charges even those who exposed themselves to all perils for Greece with a treason which was never acted, but which (as he thinks) might have been. Nay, of the Lacedaemonians themselves, he makes it doubtful whether they might have fallen in the battle or have yielded to the enemy, minimizing the proofs of their valor which were shown at Thermopylae; — and these indeed were small!

<sup>30</sup> After this, when he declares the shipwreck that befell the King's fleet, and how, an infinite mass of wealth being cast away, Aminocles the Magnesian, son of Cresines, was greatly enriched by it, having gotten an immense quantity of gold and silver; he could not so much as let this pass without snarling at it. "For this man," say she, "who had till then been none of the most fortunate, by wrecks became exceeding rich; for the misfortune he had in killing his son much afflicted his mind" (Herodotus, vii. 190). This indeed is manifest to every one, that he brought this golden treasure and this wealth cast up by the sea into his history, that he might make way for the inserting Aminocles's killing his son.

<sup>31</sup> Now Aristophanes the Boeotian wrote, that Herodotus demanded money of the Thebans but received none and that going about to discourse and reason with the young men, he was prohibited by the magistrates through their clownishness and hatred of learning; of which there is no other argument. But Herodotus bears witness to Aristophanes, whilst he charges the Thebans with some things falsely, with others ignorantly, and with others as hating them and having a quarrel with them. For he affirms that the Thessalians at first upon necessity inclined to the Persians, (Herodotus, vii. 172) in which he says the truth; and prophesying of the other Grecians that they would betray the Lacedaemonians, he added, that they would not do it willingly, but upon necessity, one city being taken after another. But he does not allow the Thebans the same plea of necessity, although they sent to Tempe five hundred men under the command of Mnamias, and to Thermopylae as many as Leonidas desired, who also alone with the Thespians stood by him, the rest leaving him after he was surrounded. But when the barbarian, having possessed himself of the avenues, was got into their confines, and Demaratus the Spartan, favoring in right of hospitality Attaginus, the chief of the oligarchy, had so wrought that he became the King's friend and familiar, whilst the other Greeks were in their ships, and none came on by land; then at last being forsaken did they accept conditions of peace, to which they were compelled by great necessity. For they had neither the sea and ships at hand, as had the Athenians; nor did they dwell far off, as the Spartans, who inhabited the most remote parts of Greece; but were not above a day and half's journey from the Persian army, whom they had already with the Spartans and Thespians alone resisted at the entrance of the straits, and were defeated.

But this writer is so equitable, that having said, "The Lacedaemonians, being alone and deserted by their allies, would perhaps have made a composition with Xerxes," he yet blames the Thebans, who were forced to the same act by the same necessity. But

when he could not wholly obliterate this most great and glorious act of the Thebans, yet went he about to deface it with a most vile imputation and suspicion, writing thus: "The confederates who had been sent returned back, obeying the commands of Leonidas; there remained only with the Lacedaemonians the Thespians and the Thebans: of these, the Thebans stayed against their wills, for Leonidas retained them as hostages; but the Thespians most willingly, as they said they would never depart from Leonidas and those that were with him" (Herodotus, vii. 222). Does he not here manifestly discover himself to have a peculiar pique and hatred against the Thebans, by the impulse of which he not only falsely and unjustly calumniated the city, but did not so much as take care to render his contradiction probable, or to conceal, at least from a few men, his being conscious of having knowingly contradicted himself? For having before said that Leonidas, perceiving his confederates not to be in good heart nor prepared to undergo danger, wished them to depart, he a little after adds that the Thebans were against their wills detained by him; whereas, if he had believed them inclined to the Persians, he should have driven them away though they had been willing to tarry. For if he thought that those who were not brisk would be useless, to what purpose was it to mix among his soldiers those that were suspected? Nor was the king of the Spartans and general of all Greece so senseless as to think that four hundred armed Thebans could be detained as hostages by his three hundred, especially the enemy being both in his front and rear. For though at first he might have taken them along with him as hostages; it is certainly probable that at last, having no regard for him, they would have gone away from him, and that Leonidas would have more feared his being encompassed by them than by the enemy. Furthermore, would not Leonidas have been ridiculous, to have sent away the other Greeks, as if by staying they should soon after have died, and to have detained the Thebans, that being himself about to die, he might keep them for the Greeks? For if he had indeed carried them along with him for hostages, or rather for slaves, he should not have kept them with those that were at the point of perishing, but have delivered them to the Greeks that went away. There remained but one cause that might be alleged for Leonidas's unwillingness to let them go, to wit, that they might die with him; and this our historian himself has taken away, writing thus of Leonidas's ambition: "Leonidas, considering these things, and desirous that this glory might redound to the Spartans alone, sent away his confederates rather for this than because they differed in their opinions." (Herodotus, vii. 220) For it had certainly been the height of folly to keep his enemies against their wills, to be partakers of that glory from which he drove away his confederates. But it is manifest from the effects, that Leonidas suspected not the Thebans of insincerity, but esteemed them to be his steadfast friends. For he marched with his army into Thebes, and at his request obtained that which was never granted to any other, to sleep within the temple of Hercules; and the next morning he related to the Thebans the vision that had appeared to him. For he imagined that he saw the most illustrious and greatest cities of Greece irregularly tossed and floating up and down on a very stormy and tempestuous sea; that Thebes, being carried above all the rest, was lifted up on high

to heaven, and suddenly after disappeared. And this indeed had a resemblance of those things which long after befell that city.

<sup>32</sup> Now Herodotus, in his narration of that fight, hath obscured also the bravest act of Leonidas, saying that they all fell in the straits near the hill (Herodotus, vii. 225). But the affair was otherwise managed. For when they perceived by night that they were encompassed by the barbarians, they marched straight to the enemies' camp, and got very near the King's pavilion, with a resolution to kill him and leave their lives about him. They came then to his tent, killing or putting to flight all they met; but when Xerxes was not found there, seeking him in that vast camp and wandering about, they were at last with much difficulty slain by the barbarians, who surrounded them on every side. What other acts and sayings of the Spartans Herodotus has omitted, we will write in the Life of Leonidas ; yet that hinders not but we may here set down also some few. Before Leonidas went forth to that war, the Spartans exhibited to him funeral spectacles, at which the fathers and mothers of those that went along with him were spectators. Leonidas himself, when one said to him, You lead very few with you to the battle, answered, There are many to die there. When his wife, at his departure, asked him what commands he had for her; he, turning to her, said, I command you to marry a good man, and bring him good children. After he was enclosed by the enemy at Thermopylae, desiring to save two that were related to him, he gave one of them a letter and sent him away; but he rejected it, saying angrily, I followed you as a soldier, not as a postman. The other he commanded to go on a message to the magistrates of Sparta; but he, answering, that is a messenger's business, took his shield, and stood up in his rank. Who would not have blamed another that should have omitted these things? But he who has collected and recorded the fart of Amasis, the coming of the thief's asses, and the giving of bottles, and many such like things, cannot seem to have omitted these gallant acts and these remarkable sayings by negligence and oversight, but as bearing ill-will and being unjust to some.

<sup>33</sup> He says that the Thebans, being at the first with the Greeks, fought compelled by necessity. (Herodotus, vii. 233) For belike not only Xerxes, but Leonidas also, had whipsters following his camp, by whom the Thebans were scourged and forced against their wills to fight. And what more ruthless libeller could there be than Herodotus, when he says that they fought upon necessity, who might have gone away and fled, and that they inclined to the Persians, whereas not one came in to help them. After this, he writes that, the rest making to the hill, the Thebans separated themselves from them, lifted up their hands to the barbarian, and coming near, cried with a most true voice, that they had favored the Persians, had given earth and water to the King, that now being forced by necessity they were come to Thermopylae, and that they were innocent of the King's wound. Having said these things, they obtained quarter; for they had the Thessalians for witnesses of all they said. Behold, how amidst the barbarians, exclamations, tumults of all sorts, flights and pursuits, their apology was heard, the witnesses examined; and the Thessalians, in the midst of those that were slain and

trodden under foot, all being done in a very narrow passage, patronized the Thebans, to wit, because the Thebans had but a little before driven away them, who were possessed of all Greece as far as, Thespieae, having conquered them in a battle, and slain their leader Lattamyas! For thus at that time stood matters between the Boeotians and the Thessalians, without any friendship or good-will. But yet how did the Thebans escape, the Thessalians helping them with their testimonies? Some of them, says he, were slain by the barbarians; many of them were by command of Xerxes marked with the royal mark, beginning with their leader Leontiades (Herodotus, vii. 233). Now the captain of the Thebans at Thermopylae was not Leontiades, but Anaxander, as both Aristophanes, out of the Commentaries of the Magistrates, and Nicander the Colophonian have taught us. Nor did any man before Herodotus know that the Thebans were stigmatized by Xerxes; for otherwise this would have been an excellent plea for them against his calumny, and this city might well have gloried in these marks, that Xerxes had punished Leonidas and Leontiades as his greatest enemies, having outraged the body of the one when he was dead, and caused the other to be tormented whilst living. But as to a writer who makes the barbarian's cruelty against Leonidas when dead a sign that he hated him most of all men when living, (Herodotus, vii. 238) and yet says that the Thebans, though favoring the Persians, were stigmatized by them at Thermopylae, and having been thus stigmatized, again cheerfully took their parts at Plataea, it seems to me that such a man — like that Hippoclidēs (See Herodotus, vi. 126–130) who gesticulating with his limbs by standing on his head on a table — would dance away the truth and say, It makes no difference to Herodotus.

<sup>34</sup> In the Eighth Book our author says, that the Greeks being frightened designed to fly from Artemisium into Greece, and that, being requested by the Euboeans to stay a little till they could dispose of their wives and families, they regarded them not, till such time as Themistocles, having taken money of them, divided it between Eurybiades and Adimantus, the captain of the Corinthians, and that then they stayed and had a sea-fight with the barbarians (Herodotus, viii. 4.) Yet Pindar, who was not a citizen of any of the confederate cities, but of one that was suspected to take part with the Medians, having made mention of Artemisium, brake forth into this exclamation: “This is the place where the sons of the Athenians laid the glorious foundation of liberty.” But Herodotus, by whom, as some will have it, Greece is honored, makes that victory a work of bribery and theft, saying that the Greeks, deceived by their captains, who had to that end taken money, fought against their wills. Nor does he here put an end to his malice. All men in a manner confess that, although the Greeks got the better at sea, they nevertheless abandoned Artemisium to the barbarians after they had received the news of the overthrow at Thermopylae. For it was to no purpose for them to stay there and keep the sea, the war being already within Thermopylae, and Xerxes having possessed himself of the avenues. But Herodotus makes the Greeks contriving to fly before they heard anything of Leonidas's death. For thus he says: “But they having been ill-treated, and especially the Athenians, half of whose ships were sorely shattered, consulted to take

their flight into Greece” (Herodotus, viii, 18). But let him be permitted so to name (or rather reproach) this retreat of theirs before the fight; but having before called it a flight, he both now styles it a flight, and will again a little after term it a flight; so bitterly does he adhere to this word “flight.” “Presently after this,” says he, “there came to the barbarians in the pinnacle a man of Hestiaea, who acquainted them with the flight of the Grecians from Artemisium. They, because the thing seemed incredible, kept the messenger in custody, and sent forth some light galleys to discover the truth” (Herodotus, viii, 23). But what is this you say? That they fled as conquered, whom the enemies after the fight could not believe to have fled, as having got much the better? Is then this a fellow fit to be believed when he writes of any man or city, who in one word deprives Greece of the victory, throws down the trophy, and pronounces the inscriptions they had set up to Diana Proseoa (EASTWARD-FACING) to be nothing but pride and vain boasting? The tenor of the inscription was as follows: —

When Athens youth had in a naval fight  
All Asia’s forces on this sea o’erthrown,  
And all the Persian army put to flight,  
Than which a greater scare was ever known,  
To show how much Diana they respected,  
This trophy to her honor they erected.

Moreover, not having described any order of the Greeks, nor told us what place every city of theirs held during the sea-fight, he says that in this retreat, which he calls their flight, the Corinthians sailed first and the Athenians last. (Herodotus, viii, 21.)

<sup>35</sup> He indeed ought not to have too much insulted over the Greeks that took part with the Persians, who, being by others thought a Thurian, reckons himself among the Halicarnassians, who, being Dorians by descent, went with their wives and children to the war against the Greeks. But he is so far from giving first an account of the straits they were in who revolted to the Persians, that, having related how the Thessalians sent to the Phocians, who were their mortal enemies, and promised to preserve their country free from all damage if they might receive from them a reward of fifty talents, he writ thus of the Phocians: “For the Phocians were the only people in these quarters who inclined not to the Persians, and that, as far as I upon due consideration can find, for no other reason but because they hated the Thessalians; for if the Thessalians had been affected to the Grecian affairs, I suppose the Phocians would have joined themselves to the Persians.” And yet, a little after he would say that thirteen cities of the Phocians were burned by the barbarians, their country laid waste, and the temple which was in Abae set on fire, and all of both sexes put to the sword, except those that by flight escaped to Parnassus (Herodotus, viii, 30–33 Compare ix. 17). Nevertheless, he puts those who suffered all extremities rather than lose their honesty in the same rank with those who most affectionately sided with the Persians. And when he could not blame the



Phocians actions, writing at his desk invented false causes and got up suspicions against them, and bids us judge them not by what they did, but by what they would have done if the Thessalians had not taken the same side, as if they had been prevented from treason because they found the place already occupied by others! Now if any one, going about to excuse the revolt of the Thessalians to the Persians, should say that they would not have done it but for the hatred they bare the Phocians, — whom when they saw joined to the Greeks, they against their inclinations followed the party of the Persians, — would not such a one be thought most shamefully to flatter, and for the sake of others to pervert the truth, by reigning good causes for evil actions? Indeed, I think, he would. Why then would not he be thought openly to calumniate, who says that the Phocians chose the best, not for the love of virtue, but because they saw the Thessalians on the contrary side? For neither does he refer this device to other authors, as he is elsewhere wont to do, but says that himself found it out by conjecture. He should therefore have produced certain arguments, by which he was persuaded that they, who did things like the best, followed the same counsels with the worst. For what he alleges of their hatreds is ridiculous. For neither did the difference between the Aeginetans and the Athenians, nor that between the Chalcidians and the Eretrians, nor yet that between the Corinthians and the Megarians, hinder them from fighting together for Greece. Nor did the Macedonians, their most bitter enemies, turn the Thessalians from their friendship with the barbarians, by joining the Persian party themselves. For the common danger did so bury their private grudges, that banishing their other passions, they applied their minds either to honesty for the sake of virtue, or to profit through the impulse of necessity. And indeed, after that necessity which compelled them to obey the Persians was over, they returned again to the Greeks, as Lacrates the Spartan has openly testified of them. And Herodotus, as constrained to it, in his relation of the affairs at Plataea, confessed that the Phocians took part with the Greeks (Herodotus, ix. 31).

<sup>36</sup> Neither ought it to seem strange to any, if he thus bitterly inveighs against the unfortunate; since he reckons amongst enemies and traitors those who were present at the engagement, and together with the other Greeks hazarded their safety. For the Naxians, says he, sent three ships to the assistance of the barbarians; but Democritus, one of their captains, persuaded the others to take the party of the Greeks (Herodotus, viii, 46). So unable he is to praise without dispraising, that if he commends one man he must condemn a whole city or people. But in this there give testimony against him, of the more ancient writers Hellanicus, and of the later Ephorus, one of which says that the Naxians came with six ships to aid the Greeks, and the other with five. And Herodotus convinces himself of having feigned these things. For the writers of the Naxian annals say, that they had before beaten back Megabates, who came to their island with two hundred ships, and after that had put to flight the general Datis who had set their city on fire. Now if, as Herodotus has elsewhere said, the barbarians burned their city so that the men were glad to save themselves by flying into the mountains, had they not just cause rather to send aid to the destroyers of their country than to help the protectors of

the common liberty? But that he framed this lie not so much to honor Democritus, as to cast infamy on the Naxians, is manifest from his omitting and wholly passing over in silence the valiant acts then performed by Democritus, of which Simonides gives us an account in this epigram: —

When as the Greeks at sea the Medes did meet,  
And had near Salamis a naval fight,  
Democritus as third led up the fleet,  
Charging the enemy with all his might;  
He took five of their ships, and did another,  
Which they had taken from the Greeks, recover.

<sup>37</sup> But why should any one be angry with him about the Naxians? If we have, as some say, antipodes inhabiting the other hemisphere, I believe that they also have heard of Themistocles and his counsel, which he gave to the Greeks, to fight a naval battle before Salamis, on which, the barbarian being overcome, he built in Melite a temple to Diana the Counsellor. This gentle writer, endeavoring, as much as in him lies, to deprive Themistocles of the glory of this, and transfer it to another, writes thus word for word: “Whilst things were thus, Mnesiphilus, an Athenian, asked Themistocles, as he was going aboard his ship, what had been resolved on in council. And being answered, that it was decreed the ships should be brought back to Isthmus, and a battle fought at sea before Peloponnesus; he said, If then they remove the navy from Salamis, you will no longer be fighting for one country for they will return every one to his own city. Wherefore, if there be any way left, go and endeavor to break this resolution; and, if it be possible, persuade Eurybiades to change his mind and stay here.” Then adding that this advice pleased Themistocles, who, without making any reply, went straight to Eurybiades, he has these very expressions: “And sitting by him he related what he had heard from Mnesiphilus, feigning as if it came from himself, and adding other things” (Herodotus, viii, 57 — 58). You see how he accuses Themistocles of disingenuity in arrogating to himself the counsel of Mnesiphilus.

<sup>38</sup> And deriding the Greeks still further, he says, that Themistocles, who was called another Ulysses for his wisdom, was so blind that he could not foresee what was fit to be done; but that Artemisia, who was of the same city with Herodotus, without being taught by any one, but by her own consideration, said thus to Xerxes: “The Greeks will not long be able to hold out against you, but you will put them to flight, and they will retire to their own cities; nor is it probable, if you march your army by land to Peloponnesus, that they will sit still, or take care to fight at sea for the Athenians. But if you make haste to give them a naval battle, I fear lest your fleets receiving damage may prove also very prejudicial to your land-forces” (Herodotus, viii, 68). Certainly Herodotus wanted nothing but verses to make Artemisia another Sibyl, so exactly prophesying of things to come. Therefore Xerxes also delivered his sons to her to be

carried to Ephesus for he had (it seems) forgot to bring women with him from Susa, if indeed the children wanted a train of female attendants.

<sup>39</sup> But it is not our design to search into the lies of Herodotus; we only make inquiry into those which he invented to detract from the glory of others. He says: "It is reported by the Athenians that Adimantus, captain of the Corinthians, when the enemies were now ready to join battle, was struck with such fear and astonishment that he fled; not thrusting his ship backward by the stern, or leisurely retreating through those that were engaged, but openly hoisting up his sails, and turning the heads of all his vessels. And about the farther part of the Salaminian coast, he was met by a pinnace, out of which one spake thus to him: Thou indeed, Adimantus, fliest, having betrayed the Grecians; yet they overcome, and according to their desires have the better of their enemies" (Herodotus, viii. 94.). This pinnace was certainly let down from heaven. For what should hinder him from erecting a tragical machine, who by his boasting excelled the tragedians in all other things? Adimantus then crediting him (he adds) "returned to the fleet, when the business was already done." "This report," says he, "is believed by the Athenians; but the Corinthians deny it, and say, they were the first at the sea-fight, for which they have the testimony of all the other Greeks." Such is this man in many other places. He spreads different calumnies and accusations of different men, that he may not fail of making some one appear altogether wicked. And it has succeeded well with him in this place; for if the calumny is believed, the Corinthians — if it is not, the Athenians — are rendered infamous. But in reality the Athenians did not belie the Corinthians, but he hath belied them both. Certainly Thucydides, bringing in an Athenian ambassador contesting with a Corinthian at Sparta, and gloriously boasting of many things about the Persian war and the sea-fight at Salamis, charges not the Corinthians with any crime of treachery or leaving their station. Nor was it likely the Athenians should object any such thing against Corinth, when they saw her engraven in the third place after the Lacedaemonians and themselves on those spoils which, being taken from the barbarians, were consecrated to the gods. And in Salamis they had permitted them to bury the dead near the city, as being men who had behaved themselves gallantly, and to write over them this elegy: —

Well-watered Corinth, stranger, was our home;  
Salamis, Ajax's isle, is now our grave;  
Here Medes and Persians and Phoenician ships  
We fought and routed, sacred Greece to save.

And their honorary sepulchre at the Isthmus has on it this epitaph: —

When Greece upon the point of danger stood,  
We fell, defending her with our life-blood.

Moreover, on the offerings of Diodorus, one of the Corinthian sea-captains, reserved in the temple of Latona, there is this inscription: —

Diodorus's seamen to Latona sent  
These arms, of hostile Medes the monument

And as for Adimantus himself, against whom Herodotus frequently inveighs, — saying, that he was the only captain who went about to fly from Artemisium, and would not stay the fight, — behold in how great honor he is: —

Here Adimantus rests: the same was he,  
Whose counsels won for Greece the crown of liberty.

For neither is it probable, that such honor would have been shown to a coward and a traitor after his decease; nor would he have dared to give his daughters the names of Nausinica, Acrothinius, and Alexibia, and his son that of Aristneas, if he had not performed some illustrious and memorable action in that fight. Nor is it credible that Herodotus was ignorant of that which could not be unknown even to the meanest Carian, that the Corinthian women alone made that glorious and divine prayer, by which they besought the Goddess Venus to inspire their husbands with a love of fighting against the barbarians. For it was a thing divulged abroad, concerning which Simonides made an epigram to be inscribed on the brazen image set up in that temple of Venus which is said to have been founded by Medea, when she desired the goddess, as some affirm, to deliver her from loving her husband Jason, or, as others say, to free him from loving Thetis. The tenor of the epigram follows: —

For those who, fighting on their country's side,  
Opposed th' imperial Mede's advancing tide,  
We, votaresses, to Cythera pray'd;  
Th' indulgent power vouchsafed her timely aid,  
And kept the citadel of Hellas free  
From rude assaults of Persia's archery.

These things he should rather have written and recorded, than have inserted Aminocles's killing of his son.

<sup>40</sup> After he had abundantly satisfied himself with the accusations brought against Themistocles, — of whom he says that, unknown to the other captains, he incessantly robbed and spoiled the islands, — (Herodotus, viii, 112) he at length openly takes away the crown of victory from the Athenians, and sets it on the head of the Aeginetans, writing thus: “The Greeks having sent the first-fruits of their spoils to Delphi, asked in general of the god, whether he had a sufficient part of the booty and were contented with it. He answered, that he had enough of all the other Greeks, but not of the Aeginetans for

he expected a donary of them, as having won the greatest honor in the battle at Salamis.” (Herodotus, viii, 122) See here how he attributes not his fictions to the Scythians, to the Persians, or to the Egyptians, as Aesop did his to the ravens and apes; but using the very person of the Pythian Apollo, he takes from Athens the chief honor of the battle at Salamis. And the second place in honor being given to Themistocles at the Isthmus by all the other captains, — every one of which attributed to himself the first degree of valor, but give the next to Themistocles, — and the judgment not coming to a determination, when he should have reprehended the ambition of the captains, he said, that all the Greeks weighed anchor from thence through envy, not being willing to give the chief honor of the victory to Themistocles (Herodotus, viii, 123 — 124).

<sup>41</sup> In his ninth and last book, having nothing left to vent his malice on but the Lacedaemonians and their glorious action against the barbarians at Plataea, he writes, that the Spartans at first feared lest the Athenians should suffer themselves to be persuaded by Mardonius to forsake the other Greeks; but that now, the Isthmus being fortified, they, supposing all to be safe at Peloponnesus, slighted the rest, feasting and making merry at home, and deluding and delaying the Athenian ambassadors. (Herodotus, ix. 8. See also viii. 141.) How then did there go forth from Sparta to Plataea a thousand and five men, having every one of them with him seven Helots? Or how came it that, exposing themselves to so many dangers, they vanquished and overthrew so many thousand barbarians? Hear now his probable cause of it. “It happened,” says he, “that there was then at Sparta a certain stranger of Tegea, named Chileus, who had some friends amongst the Ephori, between whom and him there was mutual hospitality. He then persuaded them to send forth the army, telling them that the fortification on the Isthmus, by which they had fenced in Peloponnesus, would be of no avail if the Athenians joined themselves with Mardonius” (Herodotus, ix. 9). This counsel then drew Pausanias with his army to Plataea; but if any private business had kept that Chileus at Tegea, Greece had never been victorious.

<sup>42</sup> Again, not knowing what to do with the Athenians, he tosses to and fro that city, sometimes extolling it, and sometimes debasing it. He says that, contending for the second place with the Tegeatans they made mention of the Heraclidae, alleged their acts against the Amazons, and the sepulchres of the Peloponnesians that died under the walls of Cadmea, and at last brought down their discourse to the battle of Marathon, saying, however, that they would be satisfied with the command of the left wing. (Herodotus, ix. 26, 27) A little after, he says, Pausanias and the Spartans yielded them the first place, desiring them to fight in the right wing against the Persians and give them the left, who excused themselves as not skilled in fighting against the barbarians. (Herodotus, ix. 46) Now it is a ridiculous thing, to be unwilling to fight against an enemy unless one has been used to him. But he says farther, that the other Greeks being led by their captains to encamp in another place, as soon as they were moved, the horse fled with joy towards Plataea, and in their flight came as far as Juno’s temple. (Herodotus, ix. 52) In which place indeed he charges them all in general with disobedience, cowardice, and treason.

At last he says, that only the Lacedaemonians and the Tegeates fought with the barbarians, and the Athenians with the Thebans; equally defrauding all the other cities of their part in the honor of the victory, whilst he affirms that none of them joined in the fight, but that all of them, sitting still hard by in their arms, betrayed and forsook those who fought for them; that the Phliasians and Megarians indeed, when they heard Pausanias had got the better, came in later, and falling on the Theban horse, were all cut off; that the Corinthians were not at the battle, and that after the victory, by hastening on over the hills, they escaped the Theban cavalry. (See the account of the battle of Plataea, Herodotus, ix, 59–70) For the Thebans, after the barbarians were overthrown, going before with their horse, affectionately assisted them in their flight; to return them thanks (forsooth) for the marks they had stigmatized them with at Thermopylae! Now what rank the Corinthians had in the fight at Plataea against the barbarians, and how they performed their duty, you may hear from Simonides in these verses:

I' th' midst were men, in warlike feats excelling,  
Who Ephyre full of springs, inhabited,  
And who in Corinth, Glaucus' city, dwelling,  
Great praise by their great valor merited;  
Of which they to perpetuate the fame,  
To th' gods of well-wrought gold did offerings frame.

For he wrote not these things, as one that taught at Corinth or that made verses in honor of the city, but only as recording these actions in elegiac verses. But Herodotus, whilst he desires to prevent that objection by which those might convince him of lying who should ask, Whence then are so many mounts, tombs, and monuments of the dead, at which the Plataeans, even to this day, celebrate funeral solemnities in the presence of the Greeks? — has charged, unless I am mistaken, a fouler crime than that of treason on their posterity. For these are his words: “As for the other sepulchres that are seen in Plataea, I have heard that their successors, being ashamed of their progenitors' absence from this battle, erected every man a monument for posterity's sake” (Herodotus, ix. 85). Of this treacherous deserting the battle Herodotus was the only man that ever heard. For if any Greeks withdrew themselves from the battle, they must have deceived Pausanias, Aristides, the Lacedaemonians, and the Athenians. Neither yet did the Athenians exclude the Aeginetans who were their adversaries from the inscription, nor convince the Corinthians of having fled from Salamis before the victory, Greece bearing witness to the contrary. Indeed Cleadas, a Plataean, ten years after the Persian war, to gratify, as Herodotus says, the Aeginetans, erected a mount bearing their name. Now came it then to pass that the Athenians and Lacedaemonians, who were so jealous of each other that they were presently after the war ready to go together by the ears about the setting up a trophy, did not yet repel those Greeks who fled in a fear from the battle from having a share in the honor of those that behaved themselves valiantly, but

inscribed their names on the trophies and colossuses, and granted them part of the spoils? Lastly they set up an altar, on which was engraven this epigram:

The Greeks, by valor having put to flight  
The Persians and preserved their country's right,  
Erected here this altar which you see,  
To Jove, preserver of their liberty.

Did Cleadas, O Herodotus, or some other, write this also, to oblige the cities by flattery? What need had they then to employ fruitless labor in digging up the earth, to make tombs and erect monuments for posterity's sake, when they saw their glory consecrated in the most illustrious and greatest donaries? Pausanias, indeed, when he was aspiring to the tyranny, set up this inscription in Delphi: —

Pausanias, of Greeks the general  
When he the Medes in fight had overthrown,  
Offered to Phoebus a memorial  
Of victory, this monumental stone.

In which he gave the glory to the Greeks, whose general he professed himself to be. Yet the Greeks not enduring but utterly misliking it, the Lacedaemonians, sending to Delphi, caused this to be cut out, and the names of the cities, as it was fit, to be engraven instead of it. Now how is it possible that the Greeks should have been offended that there was no mention made of them in the inscription, if they had been conscious to themselves of deserting the fight? or that the Lacedaemonians would have erased the name of their leader and general, to insert deserters and such as withdrew themselves from the common danger? For it would have been a great indignity, that Sophanes, Aeimnestus, and all the rest who showed their valor in that fight, should calmly suffer even the Cythnians and Melians to be inscribed on the trophies; and that Herodotus, attributing that fight only to three cities, should raze all the rest out of those and other sacred monuments and donaries.

<sup>43</sup> There having been then four fights with the barbarians; he says, that the Greeks fled from Artemisium; that, whilst their king and general exposed himself to danger at Thermopylae, the Lacedaemonians sat negligent at home, celebrating the Olympian and Carnean feasts; and discoursing of the action at Salamis, he uses more words about Artemisia than he does in his whole narrative of the naval battle. Lastly, he says, that the Greeks sat still at Plataea, knowing no more of the fight, till it was over, than if it had been a skirmish between mice and frogs (like that which Pigres, Artemisia's fellow countryman, merrily and scoffingly related in a poem), and it had been agreed to fight silently, lest they should be heard by others; and that the Lacedaemonians excelled not the barbarians in valor, but only got the better, as fighting against naked and unarmed

men. To wit, when Xerxes himself was present, the barbarians were with much difficulty compelled by scourges to fight with the Greeks; but at Plataea, having taken other resolutions, as Herodotus says, “they were no way inferior in courage and strength; but their garments being without armor was prejudicial to them, since being naked they fought against a completely armed enemy.” What then is there left great and memorable to the Grecians of those fights, if the Lacedaemonians fought with unarmed men, and the other Greeks, though present, were ignorant of the battle; if empty monuments are set up everywhere, and tripods and altars full of lying inscriptions are placed before the gods; if, lastly, Herodotus only knows the truth, and all others that give any account of the Greeks have been deceived by the fame of those glorious actions, as the effect of an admirable prowess? But he is an acute writer, his style is pleasant, there is a certain grace, force, and elegancy in his narrations; and he has, like a musician, elaborated his discourse, though not knowingly, still clearly and elegantly. These things delight, please, and affect all men. But as in roses we must beware of the venomous flies called cantharides; so must we take heed of the calumnies and envy lying hid under smooth and well-couched phrases and expressions, lest we imprudently entertain absurd and false opinions of the most excellent and greatest cities and men of Greece.



# On the opinions of the philosophers



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## **Book I.**

It being our determination to discourse of Natural Philosophy, we judge it necessary, in the first place and chiefly, to divide the body of philosophy into its proper members, so that we may know what is that which is called philosophy, and what part of it is physical, or the explanation of natural things. The Stoics affirm that wisdom is the knowledge of things human and divine; that philosophy is the pursuit of that art which is convenient to this knowledge; that virtue is the sole and sovereign art which is thus convenient; and this distributes itself into three general parts — natural, moral, and logical. By which just reason (they say) philosophy is tripartite; of which one natural, the other moral, the third logical. The natural when our inquiries are concerning the world and all things contained in it; the ethical is the employment of our minds in those things which concern the manners of man's life; the logical (which they also call dialectical) regulates our conversation with others in speaking. Aristotle, Theophrastus, and after them almost all the Peripatetics give the same division of philosophy. It is absolutely requisite that the complete person be contemplator of things which have a being, and the practiser of those things which are decent; and this easily appears by the following instances. If the question be proposed, whether the sun, which is so conspicuous to us, be informed of a soul or inanimate, he that makes this disquisition is the thinking man; for he proceeds no farther than to consider the nature of that thing which is proposed. Likewise, if the question be propounded, whether the world be infinite, or whether beyond the system of this world there is any real being, all these things are the objects about which the understanding of man is conversant. But if these be the questions, — what measures must be taken to compose the well-ordered life of man, what are the best methods to govern and educate children, or what are the exact rules whereby sovereigns may command and establish laws, — all these queries are proposed for the sole end of action, and the man skilled therein is the moral and practical man.

### **Chapter I.**

#### **What is Nature?**

Since we have undertaken to make a diligent search into Nature, I cannot but conclude it necessary to declare what Nature is. It is very absurd to attempt a discourse of the essence of natural things, and not to understand what is the power and sphere of Nature. If Aristotle be credited, Nature is the principle of motion and rest, in that thing in which it exists as a principle and not by accident. For all things that are conspicuous to our eyes, which are neither fortuitous nor necessary, nor have a divine original, nor acknowledge any such like cause, are called natural and enjoy their proper nature. Of

this sort are earth, fire, water, air, plants, animals; to these may be added all things produced from them, such as showers, hail, thunders, hurricanes, and winds. All these confess they had a beginning, none of these were from eternity, but had something as the origin of them; and likewise animals and plants have a principle whence they are produced. But Nature, which in all these things hath the priority, is not only the principle of motion but of repose; whatsoever enjoys the principle of motion, the same has a possibility to find a dissolution. Therefore on this account it is that Nature is the principle of motion and rest.

## **Chapter II.**

### **What is the Difference Between a Principle and an Element?**

The followers of Aristotle and Plato conclude that elements are discriminated from principles. Thales the Milesian supposeth that a principle and the elements are one and the same thing, but it is evident that they vastly differ one from another. For the elements are things compounded; but we do pronounce that principles admit not of a composition, nor are the effects of any other being. Those which we call elements are earth, water, air, and fire. But we call those principles which have nothing prior to them out of which they are produced; for otherwise not these themselves, but rather those things whereof they are produced, would be the principles. Now there are some things which have a pre-existence to earth and water, from which they are begotten; to wit, matter, which is without form or shape; then form, which we call [Greek omitted] (actuality); and lastly, privation. Thales therefore is most in error, by affirming that water is both an element and a principle.

## **Chapter III.**

### **What are Principles?**

Thales the Milesian doth affirm that water is the principle from whence all things in the universe spring. This person appears to be the first of philosophers; from him the Ionic sect took its denomination, for there are many families and successions amongst philosophers. After he had professed philosophy in Egypt, when he was very old, he returned to Miletus. He pronounced, that all things had their original from water, and into water all things are resolved. His first ground was, that whatsoever was the prolific seed of all animals was a principle, and that is moist; so that it is probable that all things receive their original from humidity. His second reason was, that all plants are nourished and fructified by that thing which is moist, of which being deprived they wither away. Thirdly, that that fire of which the sun and stars are made is nourished by watery exhalations, — yea, and the world itself; which moved Homer to sing that the generation of it was from water: —

The ocean is  
Of all things the kind genesis.  
(Iliad, xiv. 246.)

Anaximander, who himself was a Milesian, assigns the principle of all things to the Infinite, from whence all things flow, and into the same are corrupted; hence it is that infinite worlds are framed, and those dissolve again into that whence they have their origin. And thus he farther proceeds, For what other reason is there of an Infinite but this, that there may be nothing deficient as to the generation or subsistence of what is in Nature? There is his error, that he doth not acquaint us what this Infinite is, whether it be air, or water, or earth, or any other such like body. Besides he is mistaken, in that, giving us the material cause, he is silent as to the efficient cause of beings; for this thing which he makes his Infinite can be nothing but matter; but operation cannot come about in the sphere of matter, except an efficient cause be annexed. Anaximenes his fellow-citizen pronounceth, that air is the principle of all beings; from it all receive their original, and into it all return. He affirms that our soul is nothing but air; it is that which constitutes and preserves; the whole world is invested with spirit and air. For spirit and air are synonymous. This person is in this deficient, in that he concludes that of pure air, which is a simple body and is made of one only form, all animals are composed. It is not possible to think that a single principle should be the matter of all things, from whence they receive their subsistence; besides this there must be an operating cause. Silver (for example) is not of itself sufficient to frame a drinking cup; an operator also is required, which is the silversmith. The like may be applied to vessels made of wood, brass, or any other material.

Anaxagoras the Clazomenian asserted Homoeomeries (or parts similar or homogeneous) to be the original cause of all beings; it seemed to him impossible that anything could arise of nothing or be dissolved into nothing. Let us therefore instance in nourishment, which appears simple and uniform, such as bread which we owe to Ceres and water which we drink. Of this very nutriment, our hair, our veins, our arteries, nerves, bones, and all our other parts are nourished. These things thus being performed, it must be granted that the nourishment which is received by us contains all those things by which these of us are produced. In it there are those particles which are producers of blood, bones, nerves, and all other parts; these particles (he thought) reason discovers for us. For it is not necessary that we should reduce all things under the objects of sense; for bread and water are fitted to the senses, yet in them there are those particles latent which are discoverable only by reason. It being therefore plain that there are particles in the nourishment similar to what is produced by it, he terms these homogeneous parts, averring that they are the principles of beings. Matter is according to him these similar parts, and the efficient cause is a Mind, which orders all things that have an existence. Thus he begins his discourse: "All things were confused one among

another; but Mind divided and brought them to order.” In this he is to be commended, that he yokes together matter and an intellectual agent.

Archelaus the son of Apollodorus, the Athenian, pronounceth, that the principles of all things have their origin from an infinite air rarefied or condensed. Air rarefied is fire, condensed is water.

These philosophers, the followers of Thales, succeeding one another, made up that sect which takes to itself the denomination of the Ionic.

Pythagoras the Samian, the son of Mnesarchus, from another origin deduces the principles of all things; it was he who first called philosophy by its name. He thought the first principles to be numbers, and those symmetries in them which he styles harmonies; and the composition of both he terms elements, called geometrical. Again, he places unity and the indefinite binary number amongst the principles. One of these principles ends in an efficient and forming cause, which is Mind, and that is God; the other to the passive and material part, and that is the visible world. Moreover, the nature of number (he saith) consists in the ten; for all people, whether Grecians or barbarians, reckon from one to ten, and thence return to one again. Farther he avers the virtue of ten consists in the quaternion; the reason whereof is this, — if any person start from one, and add numbers so as to take in the quaternary, he shall complete the number ten; if he passes the four, he shall go beyond the ten; for one, two, three, and four being added up together make ten. The nature of numbers, therefore, if we regard the units, abideth in the ten; but if we regard its power, in the four. Therefore the Pythagoreans say that their most sacred oath is by that god who delivered to them the quaternary.

By th’ founder of the sacred number four,  
Eternal Nature’s font and source, they swore.

Of this number the soul of man is composed; for mind, knowledge, opinion, and sense are the four that complete the soul, from which all sciences, all arts, all rational faculties derive themselves. For what our mind perceives, it perceives after the manner of a thing that is one, the soul itself being a unity; as for instance, a multitude of persons are not the object of our sense nor are comprehended by us, for they are infinite; our understanding gives the general concept of A MAN, in which all individuals agree. The number of individuals is infinite; the generic or specific nature of all being is a unit, or to be apprehended as one only thing; from this one conception we give the genuine measures of all existence, and therefore we affirm that a certain class of beings are rational and discursive. But when we come to give the nature of a horse, it is that animal which neighs; and this being common to all horses, it is manifest that the understanding, which hath such like conceptions, is in its nature unity. It follows that the number called the infinite binary must be science; in every demonstration or belief belonging to science, and in every syllogism, we draw that conclusion which is in dispute from those propositions which are by all granted, by which means another

proposition is obtained from the premises. The comprehension of these we call knowledge; for which reason science is the binary number. But opinion is the ternary; for that rationally follows from comprehension. The objects of opinion are many things, and the ternary number denotes a multitude, as “Thrice happy Grecians”; for which reason Pythagoras admits the ternary. This sect of philosophers is called the Italic, by reason Pythagoras started his school in Italy; his hatred of the tyranny of Polycrates enforced him to abandon his native country Samos. Heraclitus and Hippasus of Metapontum suppose that fire gives the origination to all beings, that they all flow from fire, and in fire they all conclude; for of fire when first quenched the world was constituted. The first part of the world, being most condensed and contracted within itself, made the earth; but part of that earth being loosened and made thin by fire, water was produced; afterwards this water being exhaled and rarefied into vapors became air; after all this the world itself, and all other corporeal beings, shall be dissolved by fire in the universal conflagration. By them therefore it appears that fire is what gives beginning to all things, and is that in which all things receive their period.

Epicurus the son of Neocles, the Athenian, his philosophical sentiments being the same with those of Democritus, affirms that the principles of all being are bodies which are only perceptible by reason; they admit not of a vacuity, nor of any original, but being of a self-existence are eternal and incorruptible; they are not liable to any diminution, they are indestructible, nor is it possible for them to receive any transformation of parts, or admit of any alterations; of these reason is only the discoverer; they are in a perpetual motion in vacuity, and by means of the empty space; for the vacuum itself is infinite, and the bodies that move in it are infinite. Those bodies acknowledge these three accidents, figure, magnitude, and gravity. Democritus acknowledged but two, magnitude and figure. Epicurus added the third, to wit, gravity; for he pronounced that it is necessary that bodies receive their motion from that impression which springs from gravity, otherwise they could not be moved. The figures of atoms cannot be incomprehensible, but they are not infinite. These figures are neither hooked nor trident-shaped nor ring-shaped, such figures as these being exposed to collision; but the atoms are impassible, impenetrable; they have indeed figures of their own, which are conceived only by reason. It is called an atom, by reason not of its smallness but of its indivisibility; in it no vacuity, no passible affection is to be found. And that there is an atom is perfectly clear; for there are elements which have a perpetual duration, and there are animals which admit of a vacuity, and there is a unity.

Empedocles the Agrigentine, the son of Meton, affirms that there are four elements, fire, air, earth, and water, and two powers which bear the greatest command in nature, concord and discord, of which one is the union, the other the division of beings. Thus he sings,

Hear first the four roots of all created things: —  
Bright shining Jove, Juno that beareth life,

Pluto beneath the earth, and Nestis who  
Doth with her tears water the human fount.

By Jupiter he understands fire and ether, by Juno that gives life he means the air, by Pluto the earth, by Nestis and the spring of all mortals (as it were) seed and water.

Socrates the son of Sophroniscus, and Plato son of Ariston, both natives of Athens, entertain the same opinion concerning the universe; for they suppose three principles, God, matter, and the idea. God is the universal understanding; matter is that which is the first substratum, accommodated for the generation and corruption of beings; the idea is an incorporeal essence, existing in the cogitations and apprehensions of God; for God is the soul and mind of the world.

Aristotle the son of Nichomachus, the Stagirite, constitutes three principles; Entelecheia (which is the same with form), matter, and privation. He acknowledges four elements, and adds a certain fifth body, which is ethereal and not obnoxious to mutation.

Zeno son of Mnaseas, the native of Citium, avers these to be principles, God and matter, the first of which is the efficient cause, the other the passible and receptive. Four more elements he likewise confesses.

## **Chapter IV.**

### **How Was This World Composed in that Order and After that Manner it is?**

The world being broken and confused, after this manner it was reduced into figure and composure as now it is. The insectible bodies or atoms, by a wild and fortuitous motion, without any governing power, incessantly and swiftly were hurried one amongst another, many bodies being jumbled together; upon this account they have a diversity in the figures and magnitude. These therefore being so jumbled together, those bodies which were the greatest and heaviest sank into the lowest place; they that were of a lesser magnitude, being round, smooth, and slippery, these meeting with those heavier bodies were easily broken into pieces, and were carried into higher places. But when that force whereby these variously figured particles fought with and struck one another, and forced the lighter upwards, did cease, and there was no farther power left to drive them into superior regions, yet they were wholly hindered from descending downwards, and were compelled to reside in those places capable to receive them; and these were the heavenly spaces, unto which a multitude of these small bodies were hurled, and these being thus shivered fell into coherence and mutual embraces, and by this means the heaven was produced. Then a various and great multitude of atoms enjoying the same nature, as it is before asserted, being hurried aloft, did form the stars. The multitude of these exhaled bodies, having struck and broke the air in shivers, forced a passage through it; this being turned into wind invested the stars, as it moved, and

whirled them about, by which means to this present time that circular motion which these stars have in the heavens is maintained. Much after the same manner the earth was made; for by those little particles whose gravity made them to reside in the lower places the earth was formed. The heaven, fire, and air were constituted of those particles which were carried aloft. But a great deal of matter remaining in the earth, this being condensed by the driving of the winds and the air from the stars, every little part and form of it was compressed, which created the element of water; but this being fluidly disposed did run into those places which were hollow, and these places were those that were capable to receive and protect it; or the water, subsisting by itself, did make the lower places hollow. After this manner the principal parts of the world were constituted.

## **Chapter V.**

### **Whether the Universe is One Single Thing.**

The Stoics pronounce that the world is one thing, and this they say is the universe and is corporeal.

But Empedocles's opinion is, that the world is one; yet by no means the system of this world must be styled the universe, but that it is a small part of it, and the remainder is inactive matter.

What to Plato seems the truest he thus declares, that there is one world, and that world is the universe; and this he endeavors to evince by three arguments. First, that the world could not be complete and perfect, if it did not within itself include all beings. Secondly, nor could it give the true resemblance of its original and exemplar, if it were not the one only begotten thing. Thirdly, it could not be incorruptible, if there were any being out of its compass to whose power it might be obnoxious. But to Plato it may be thus returned. First, that the world is not complete and perfect, nor doth it contain all things within itself. And if man is a perfect being, yet he doth not encompass all things. Secondly, that there are many exemplars and originals of statues, houses, and pictures. Thirdly, how is the world perfect, if anything beyond it is possible to be moved about it? But the world is not incorruptible, nor can it be so conceived, because it had an original. To Metrodorus it seems absurd, that in a large field one only stalk should grow, and in an infinite space one only world exist; and that this universe is infinite is manifest by this, that there is an infinity of causes. Now if this world be finite and the causes producing it infinite, it follows that the worlds likewise be infinite; for where all causes concur, there the effects also must appear, let the causes be what they will, either atoms or elements.

## **Chapter VI.**



## Whence Did Men Obtain the Knowledge of the Existence and Essence of a Deity?

The Stoics thus define the essence of a god. It is a spirit intellectual and fiery, which acknowledges no shape, but is continually changed into what it pleases, and assimilates itself to all things. The knowledge of this deity they first received from the pulchritude of those things which so visibly appeared to us; for they concluded that nothing beauteous could casually or fortuitously be formed, but that it was framed from the art of a great understanding that produced the world. That the world is very resplendent is made perspicuous from the figure, the color, the magnitude of it, and likewise from the wonderful variety of those stars which adorn this world. The world is spherical; the orbicular hath the pre-eminence above all other figures, for being round itself it hath its parts like itself. (On this account, according to Plato, the understanding, which is the most sacred part of man, is in the head.) The color of it is most beauteous; for it is painted with blue; which, though little blacker than purple, yet hath such a shining quality, that by reason of the vehement efficacy of its color it cuts through such a space of air; whence it is that at so great a distance the heavens are to be contemplated. And in this very greatness of the world the beauty of it appears. View all things: that which contains the rest carries a beauty with it, as an animal or a tree. Also things which are visible to us accomplish the beauty of the world. The oblique circle called the Zodiac in heaven is with different images painted and distinguished: —

There's Cancer, Leo, Virgo, and the Claws;  
Scorpio, Arcitenens, and Capricorn;  
Amphora, Pisces, then the Ram, and Bull;  
The lovely pair of Brothers next succeed.  
(From Aratus.)

There are a thousand others that give us the suitable reflections of the beauty of the world. Thus Euripides: —

The starry splendor of the skies,  
The beautiful and varied work of that wise  
Creator, Time.

From this the knowledge of a god is conveyed to man; that the sun, the moon, and the rest of the stars, being carried under the earth, rise again in their proper color, magnitude, place, and times. Therefore they who by tradition delivered to us the knowledge and veneration of the gods did it by these three manner of ways: — first, from Nature; secondly, from fables; thirdly, from the testimony supplied by the laws of commonwealths. Philosophers taught the natural way; poets, the fabulous; and the political way is to be had from the constitutions of each commonwealth. All sorts of this

learning are distinguished into these seven parts. The first is from things that are conspicuous, and the observation of those bodies which are in places superior to us. To men the heavenly bodies that are so visible did give the knowledge of the deity; when they contemplated that they are the causes of so great an harmony, that they regulate day and night, winter and summer, by their rising and setting, and likewise considered those things which by their influences in the earth do receive a being and do likewise fructify. It was manifest to men that the Heaven was the father of those things, and the Earth the mother; that the Heaven was the father is clear, since from the heavens there is the pouring down of waters, which have their spermatic faculty; the Earth the mother, because she receives them and brings forth. Likewise men considering that the stars are running (Greek omitted) in a perpetual motion, that the sun and moon give us the stimulus to view and contemplate (Greek omitted), they call them all gods (Greek omitted). In the second and third place, they thus distinguished the deities into those which are beneficial and those that are injurious to mankind. Those which are beneficial they call Jupiter, Juno, Mercury, Ceres; those who are mischievous the Dirae, Furies, and Mars. These, which threaten dangers and violence, men endeavor to appease and conciliate by sacred rites. The fourth and the fifth order of gods they assign to things and passions; to passions, Love, Venus, and Desire; the deities that preside over things, Hope, Justice, and Eunomia.

The sixth order of deities are the ones made by the poets; Hesiod, willing to find out a father for those gods that acknowledge an original, invented their progenitors, —

Hyperion, Coeus, and Iapetus,  
With Creius:  
(Hesiod, "Theogony," 134.)

upon which account this is called the fabulous. The seventh rank of the deities added to the rest are those which, by their beneficence to mankind, were honored with a divine worship, though they were born of mortal race; of this sort were Hercules, Castor and Pollux, and Bacchus. These are reputed to be of a human species; for of all beings that which is divine is most excellent, and man amongst all animals is adorned with the greatest beauty, is also the best, being adorned by virtue above the rest because of the gift of intellect: therefore it was thought that those who were admirable for excellence should resemble that which is the best and most beautiful.

## **Chapter VII.**

### **What is God?**

Some of the philosophers, such as Diagoras the Melian, Theodorus the Cyrenean, and Euemerus the Tegeatan, did deny unanimously that there were any gods; and

Callimachus the Cyrenean discovered his mind concerning Euemerus in these Iambic verses, thus writing: —

To th' ante-mural temple flock apace,  
Where he that long ago composed of brass  
Great Jupiter, Thrasonic old bald pate,  
Now scribbles impious books, — a boastful ass!

meaning books which prove there are no gods. Euripides the tragedian durst not openly declare his sentiment; the court of Areopagus terrified him. Yet he sufficiently manifested his thoughts by this method. He presented in his tragedy Sisyphus, the first and great patron of this opinion, and introduced himself as one agreeing with him: —

Disorder in those days did domineer,  
And brutal power kept the world in fear.

Afterwards by the sanction of laws wickedness was suppressed; but by reason that laws could prohibit only public villanies, yet could not hinder many persons from acting secret impieties, some wise persons gave this advice, that we ought to blind truth with lying disguises, and persuade men that there is a God: —

There's an eternal God does hear and see  
And understand every impiety;  
Though it in dark recess or thought committed be.

But this poetical fable ought to be rejected, he thought, along with Callimachus, who thus saith: —

If you believe a God, it must be meant  
That you conceive this God omnipotent.

But God cannot do everything; for, if it were so, then a God could make snow black, and the fire cold, and him that is in a posture of sitting to be upright, and so on the contrary. The brave-speaking Plato pronounceth that God formed the world after his own image; but this smells rank of the old dotages, old comic writers would say; for how did God, casting his eye upon himself, frame this universe? Or how can God be spherical, and be inferior to man?

Anaxagoras avers that bodies did consist from all eternity, but the divine intellect did reduce them into their proper orders, and effected the origination of all beings. But Plato did not suppose that the primary bodies had their consistence and repose, but that they were moved confusedly and in disorder; but God, knowing that order was better than confusion, did digest them into the best methods. Both these were equally peccant;

for both suppose God to be the great moderator of human affairs and for that cause to have formed this present world; when it is apparent that an immortal and blessed being, replenished with all his glorious excellencies, and not at all obnoxious to any sort of evil, but being wholly occupied with his own felicity and immortality, would not employ himself with the concerns of men; for certainly miserable is the being which, like a laborer or artificer, is molested by the troubles and cares which the forming and governing of this world must give him. Add to this, that the God whom these men profess was either not at all existing before this present world (when bodies were either reposed or in a disordered motion), or that at that time God did either sleep, or else was in a constant watchfulness, or that he did neither of these. Now neither the first nor the second can be entertained, because they suppose God to be eternal; if God from eternity was in a continual sleep, he was in an eternal death, — and what is death but an eternal sleep? — but no sleep can affect a deity, for the immortality of God and alliance to death are vastly different. But if God was in a continual vigilance, either there was something wanting to make him happy, or else his beatitude was perfectly complete; but according to neither of these can God be said to be blessed; not according to the first, for if there be any deficiency there is no perfect bliss; not according to the second, for, if there be nothing wanting to the felicity of God, it must be a needless enterprise for him to busy himself in human affairs. And how can it be supposed that God administers by his own providence human concerns, when to vain and trifling persons prosperous things happen, to great and high adverse? Agamemnon was both

A virtuous prince, for warlike acts renowned,  
("Iliad," iii. 179.)

and by an adulterer and adulteress was vanquished and perfidiously slain. Hercules, after he had freed the life of man from many things that were pernicious to it, perished by the witchcraft and poison of Deianira.

Thales said that the intelligence of the world was God.

Anaximander concluded that the stars were heavenly deities.

Democritus said that God, being a globe of fire, is the intelligence and the soul of the world.

Pythagoras says that, of his principles, unity is God; and the good, which is indeed the nature of a unity, is mind itself; but the binary number, which is infinite, is a daemon, and evil, — about which the multitude of material beings and this visible world are related.

Socrates and Plato agree that God is that which is one, hath its original from its own self, is of a singular subsistence, is one only being perfectly good; all these various names signifying goodness do all centre in mind; hence God is to be understood as that mind and intellect, which is a separate idea, that is to say, pure and unmixed of all matter, and not mingled with anything subject to passions.

Aristotle's sentiment is, that God hath his residence in superior regions, and hath placed his throne in the sphere of the universe, and is a separate idea; which sphere is an ethereal body, which is by him styled the fifth essence or quintessence. For there is a division of the universe into spheres, which are contiguous by their nature but appear to reason to be separated; and he concludes that each of the spheres is an animal, composed of a body and soul; the body of them is ethereal, moved orbicularly, the soul is the rational form, which is unmoved, and yet is the cause that the sphere is in motion.

The Stoics affirm that God is a thing more common and obvious, and is a mechanic fire which every way spreads itself to produce the world; it contains in itself all seminal virtues, and by this means all things by a fatal necessity were produced. This spirit, passing through the whole world, received different names from the mutations in the matter through which it ran in its journey. God therefore is the world, the stars, the earth, and (highest of all) the mind in the heavens. In the judgment of Epicurus all the gods are anthropomorphites, or have the shape of men; but they are perceptible only by reason, for their nature admits of no other manner of being apprehended, their parts being so small and fine that they give no corporeal representations. The same Epicurus asserts that there are four other natural beings which are immortal: of this sort are atoms, the vacuum, the infinite, and the similar parts; and these last are called Homoeomeries and likewise elements.

## **Chapter VIII.**

### **Of Those that are Called Geniuses and Heroes**

Having treated of the essence of the deities in a just order, it follows that we discourse of daemons and heroes. Thales, Pythagoras, Plato, and the Stoics do conclude that daemons are essences endowed with souls; that the heroes are the souls separated from their bodies, some are good, some are bad; the good are those whose souls are good, the evil those whose souls are wicked. All this is rejected by Epicurus.

## **Chapter IX.**

### **Of Matter.**

Matter is that first being which is substrate for generation, corruption, and all other alterations.

The disciples of Thales and Pythagoras, with the Stoics, are of opinion that matter is changeable, mutable, convertible, and sliding through all things.

The followers of Democritus aver that the vacuum, the atom, and the incorporeal substance are the first beings, and not obnoxious to passions.

Aristotle and Plato affirm that matter is of that species which is corporeal, void of any form, species, figure, and quality, but apt to receive all forms, that she may be the

nurse, the mother, and origin of all other beings. But they that do say that water, earth, air, and fire are matter do likewise say that matter cannot be without form, but conclude it is a body; but they that say that individual particles and atoms are matter do say that matter is without form.

## **Chapter X.**

### **Of Ideas.**

An idea is a being incorporeal, not subsisting by itself, but gives figure unto shapeless matter, and becomes the cause of its phenomena.

Socrates and Plato conjecture that these ideas are beings separate from matter, subsisting in the understanding and imagination of the deity, that is, of mind.

Aristotle accepted forms and ideas; but he doth not believe them separated from matter, or patterns of the things God has made.

Those Stoics, that are of the school of Zeno, profess that ideas are nothing else but the conceptions of our own mind.

## **Chapter XI.**

### **Of Causes.**

A cause is that by which anything is produced, or by which anything is effected.

Plato gives this triple division of causes, — the material, the efficient, and the final cause; the principal cause he judges to be the efficient, which is the mind and intellect.

Pythagoras and Aristotle judge the first causes are incorporeal beings, but those that are causes by accident or participation become corporeal substances; by this means the world is corporeal.

The Stoics grant that all causes are corporeal, inasmuch as they are physical.

## **Chapter XII.**

### **Of Bodies.**

A body is that being which hath these three dimensions, breadth, depth, and length; — or a bulk which makes a sensible resistance; — or whatsoever of its own nature possesseth a place.

Plato saith that it is neither heavy nor light in its own nature, when it exists in its own place; but being in the place where another should be, then it has an inclination by which it tends to gravity or levity.

Aristotle saith that, if we simply consider things in their own nature, the earth only is to be judged heavy, and fire light; but air and water are on occasions heavy and at other

times light.

The Stoics think that of the four elements two are light, fire and air; two ponderous, earth and water; that which is naturally light doth by its own nature, not by any inclination, recede from its own centre; but that which is heavy doth by its own nature tend to its centre; for the centre is not a heavy thing in itself.

Epicurus thinks that bodies are not limited; but the first bodies, which are simple bodies, and all those composed of them, all acknowledge gravity; that all atoms are moved, some perpendicularly, some obliquely; some are carried aloft either by immediate impulse or with vibrations.

### **Chapter XIII.**

#### **Of Those Things that are Least in Nature.**

Empedocles, before the four elements, introduceth the most minute bodies which resemble elements; but they did exist before the elements, having similar parts and orbicular.

Heraclitus brings in the smallest fragments, and those indivisible.

### **Chapter XIV.**

#### **Of Figures.**

A figure is the exterior appearance, the circumscription, and the boundary of a body.

The Pythagoreans say that the bodies of the four elements are spherical, fire being in the supremest place only excepted, whose figure is conical.

### **Chapter XV.**

#### **Of Colors.**

Color is the visible quality of a body.

The Pythagoreans called color the external appearance of a body. Empedocles, that which is consentaneous to the passages of the eye. Plato, that they are fires emitted from bodies, which have parts harmonious for the sight. Zeno the Stoic, that colors are the first figurations of matter. The Pythagoreans, that colors are of four sorts, white and black, red and pale; and they derive the variety of colors from the mixtures of the elements, and that seen in animals also from the variety of food and the air.

### **Chapter XVI.**

#### **Of the Division of Bodies.**

The disciples of Thales and Pythagoras grant that all bodies are passible and divisible into infinity. Others hold that atoms and indivisible parts are there fixed, and admit not of a division into infinity. Aristotle, that all bodies are potentially but not actually divisible into infinity.

## **Chapter XVII.**

### **How Bodies are Mixed and Contempered One with Another.**

The ancient philosophers held that the mixture of elements proceeded from the alteration of qualities; but the disciples of Anaxagoras and Democritus say it is done by apposition. Empedocles composes the elements of still minuter bulks, those which are the most minute and may be termed the element of elements. Plato assigns three bodies (but he will not allow these to be elements, nor properly so called), air, fire, and water, which are mutable into one another; but the earth is mutable into none of these.

## **Chapter XVIII.**

### **Of a Vacuum.**

All the natural philosophers from Thales to Plato rejected a vacuum. Empedocles says that there is nothing of a vacuity in Nature, nor anything superabundant. Leucippus, Democritus, Demetrius, Metrodorus, Epicurus, that the atoms are in number infinite; and that a vacuum is infinite in magnitude. The Stoics, that within the compass of the world there is no vacuum, but beyond it the vacuum is infinite. Aristotle, that the vacuum beyond the world is so great that the heaven has liberty to breathe into it, for the heaven is fiery.

## **Chapter XIX.**

### **Of Place.**

Plato, to define place, calls it that thing which in its own bosom receives forms and ideas; by which metaphor he denotes matter, being (as it were) a nurse or receptacle of beings. Aristotle, that it is the ultimate superficies of the circumambient body, contiguous to that which it doth encompass.

## **Chapter XX.**

### **Of Space.**



The Stoics and Epicureans make a place, a vacuum, and space to differ. A vacuum is that which is void of anything that may be called a body; place is that which is possessed by a body; a space that which is partly filled with a body, as a cask with wine.

## **Chapter XXI.**

### **Of Time.**

In the sense of Pythagoras, time is that sphere which encompasses the world. Plato says that it is a movable image of eternity, or the interval of the world's motion.

Eratosthenes, that it is the solar motion.

## **Chapter XXII.**

### **Of the Substance and Nature of Time.**

Plato says that the heavenly motion is time. Most of the Stoics that motion is time. Most philosophers think that time had no commencement; Plato, that time had only in intelligence a beginning.

## **Chapter XXIII.**

### **Of Motion.**

Plato and Pythagoras say that motion is a difference and alteration in matter. Aristotle, that it is the actual operation of that which may be moved. Democritus, that there is but one sort of motion, and it is that which is vibratory. Epicurus, that there are two species of motion, one perpendicular, and the other oblique. Herophilus, that one species of motion is obvious only to reason, the other to sense. Heraclitus utterly denies that there is anything of quiet or repose in nature; for that is the state of the dead; one sort of motion is eternal, which he assigns to beings eternal, the other perishable, to those things which are perishable.

## **Chapter XXIV.**

### **Of Generation and Corruption.**

Parmenides Melissus, and Zeno deny that there are any such things as generation and corruption, for they suppose that the universe is unmovable. Empedocles, Epicurus, and other philosophers that combine in this, that the world is framed of small corporeal particles meeting together, affirm that corruption and generation are not so properly to

be accepted; but there are conjunctions and separations, which do not consist in any distinction according to their qualities, but are made according to quantity by coalition or disjunction. Pythagoras, and all those who take for granted that matter is subject to mutation, say that generation and corruption are to be accepted in their proper sense, and that they are accomplished by the alteration, mutation, and dissolution of elements.

### **Chapter XXV.**

#### **Of Necessity.**

Thales says that necessity is omnipotent, and that it exerciseth an empire over everything. Pythagoras, that the world is invested by necessity. Parmenides and Democritus, that there is nothing in the world but what is necessary, and that this same necessity is otherwise called fate, justice, providence, and the architect of the world.

### **Chapter XXVI.**

#### **Of the Nature of Necessity.**

But Plato distinguisheth and refers some things to Providence, others to necessity. Empedocles makes the nature of necessity to be that cause which employs principles and elements. Democritus makes it to be a resistance, impulse, and force of matter. Plato sometimes says that necessity is matter; at other times, that it is the habitude or respect of the efficient cause towards matter.

### **Chapter XXVII.**

#### **Of Destiny or Fate.**

Heraclitus, who attributes all things to fate, makes necessity to be the same thing with it. Plato admits of a necessity in the minds and the acts of men, but yet he introduceth a cause which flows from ourselves. The Stoics, in this agreeing with Plato, say that necessity is a cause invincible and violent; that fate is the ordered complication of causes, in which there is an intexture of those things which proceed from our own determination, so that certain things are to be attributed to fate, others not.

### **Chapter XXVIII.**

#### **Of the Nature of Fate.**

According to Heraclitus, the essence of fate is a certain reason which penetrates the substance of all being; and this is an ethereal body, containing in itself that seminal

faculty which gives an original to every being in the universe. Plato affirms that it is the eternal reason and the eternal law of the nature of the world. Chrysippus, that it is a spiritual faculty, which in due order doth manage and rule the universe. Again, in his book styled the “Definitions,” that fate is the reason of the world, or that it is that law whereby Providence rules and administers everything that is in the world; or it is that reason by which all things have been, all things are, and all things will be produced. The Stoics say that it is a chain of causes, that is, it is an order and connection of causes which cannot be resisted. Posidonius, that it is a being the third in degree from Jupiter; the first of beings is Jupiter, the second Nature, and the third Fate.

## **Chapter XXIX.**

### **Of Fortune.**

Plato says, that it is an accidental cause and a casual consequence in things which proceed from the election and counsel of men. Aristotle, that it is an accidental cause in those things done by an impulse for a certain end; and this cause is uncertain and unstable: there is a great deal of difference betwixt that which flows from chance and that which falls out by Fortune; for that which is fortuitous allows also chance, and belongs to things practical; but what is by chance cannot be also by Fortune, for it belongs to things without action: Fortune, moreover, pertains to rational beings, but chance to rational and irrational beings alike, and even to inanimate things. Epicurus, that it is a cause not always consistent, but various as to persons, times, and manners. Anaxagoras and the Stoics, that it is that cause which human reason cannot comprehend; for there are some things which proceed from necessity, some things from Fate, some from choice and free-will, some from Fortune, some from chance.

## **Chapter XXX.**

### **Of Nature.**

Empedocles affirms that Nature is nothing else but the mixture and separation of the elements; for thus he writes in the first book of his natural philosophy: —

Nature gives neither life nor death,  
Mutation makes us die or breathe.  
The elements first are mixed, then each  
Do part: this Nature is in mortal speech.

Anaxagoras is of the same opinion, that Nature is coalition and separation, that is, generation and corruption.

## **Book II.**

Having finished my dissertation concerning principles and elements and those things which chiefly appertain to them, I will turn my pen to discourse of those things which are produced by them, and will take my beginning from the world, which contains and encompasseth all beings.

### **Chapter I.**

#### **Of the World.**

Pythagoras was the first philosopher that called the world [Greek omitted], from the order and beauty of it; for so that word signifies. Thales and his followers say the world is one. Democritus, Epicurus, and their scholar Metrodorus affirm that there are infinite worlds in an infinite space, for that infinite vacuum in its whole extent contains them. Empedocles, that the circle which the sun makes in its motion circumscribes the world, and that circle is the utmost bound of the world. Seleucus, that the world knows no limits. Diogenes, that the universe is infinite, but this world is finite. The Stoics make a difference between that which is called the universe, and that which is called the whole world; — the universe is the infinite space considered with the vacuum, the vacuity being removed gives the right conception of the world; so that the universe and the world are not the same thing.

### **Chapter II.**

#### **Of the Figure of the World.**

The Stoics say that the figure of the world is spherical, others that it is conical, others oval. Epicurus, that the figure of the world may be globular, or that it may admit of other shapes.

### **Chapter III.**

#### **Whether the World be an Animal.**

Democritus, Epicurus, and those philosophers who introduced atoms and a vacuum, affirm that the world is not an animal, nor governed by any wise Providence, but that it is managed by a nature which is void of reason. All the other philosophers affirm that the world is informed with a soul, and governed by reason and Providence. Aristotle is excepted, who is somewhat different; he is of opinion, that the whole world is not acted by a soul in every part of it, nor hath it any sensitive, rational, or intellectual faculties,

nor is it directed by reason and Providence in every part of it; of all which the heavenly bodies are made partakers, for the circumambient spheres are animated and are living beings; but those things which are about the earth are void of those endowments; and though those terrestrial bodies are of an orderly disposition, yet that is casual and not primogenial.

## **Chapter IV.**

### **Whether the World is Eternal and Incorruptible.**

Pythagoras [and Plato], agreeing with the Stoics, affirm that the world was framed by God, and being corporeal is obvious to the senses, and in its own nature is obnoxious to destruction; but it shall never perish, it being preserved by the providence of God. Epicurus, that the world had a beginning, and so shall have an end, as plants and animals have. Xenophanes, that the world never had a beginning, is eternal and incorruptible. Aristotle, that the part of the world which is sublunary is subject to change, and there terrestrial beings find a decay.

## **Chapter V.**

### **Whence Does the World Receive its Nutriment?**

Aristotle says that, if the world be nourished, it will likewise be dissolved; but it requires no aliment, and will therefore be eternal. Plato, that this very world prepares for itself a nutriment, by the alteration of those things which are corruptible in it. Philolaus affirms that a destruction happens to the world in two ways; either by fire failing from heaven, or by the sublunary water being poured down through the whirling of the air; and the exhalations proceeding from thence are aliment of the world.

## **Chapter VI.**

### **From what Element God Did Begin to Raise the Fabric of the World.**

The natural philosophers pronounce that the forming of this world took its original from the earth, it being its centre, for the centre is the principal part of the globe. Pythagoras, from the fire and the fifth element. Empedocles determines, that the first and principal element distinct from the rest was the aether, then fire, after that the earth, which earth being strongly compacted by the force of a potent revolution, water springs from it, the exhalations of which water produce the air; the heaven took its origin from the aether, and fire gave a being to the sun; those things nearest to the earth are condensed from the remainders. Plato, that the visible world was framed after the exemplar of the intellectual world; the soul of the visible world was first produced, then the corporeal

figure, first that which proceeded from fire and earth, then that which came from air and water. Pythagoras, that the world was formed of five solid figures which are called mathematical; the earth was produced by the cube, the fire by the pyramid, the air by the octahedron, the water by the icosahedron, and the globe of the universe by the dodecahedron. In all these Plato hath the same sentiments with Pythagoras.

## **Chapter VII.**

### **In what Form and Order the World Was Composed.**

Parmenides maintains that there are small coronets alternately twisted one within another, some made up of a thin, others of a condensed, matter; and there are others between mixed mutually together of light and of darkness, and around them all there is a solid substance, which like a firm wall surrounds these coronets. Leucippus and Democritus cover the world round about, as with a garment and membrane. Epicurus says that that which abounds some worlds is thin, and that which limits others is gross and condensed; and of these spheres some are in motion, others are fixed. Plato, that fire takes the first place in the world, the second the aether, after that the air, under that the water; the last place the earth possesseth: sometimes he puts the aether and the fire in the same place. Aristotle gives the first place to the aether, as that which is impassible, it being a kind of a fifth body after which he placeth those that are passible, fire, air, and water, and last of all the earth. To those bodies that are accounted celestial he assigns a motion that is circular, but to those that are seated under them, if they be light bodies, an ascending, if heavy, a descending motion. Empedocles, that the places of the elements are not always fixed and determined, but they all succeed one another in their respective stations.

## **Chapter VIII.**

### **What is the Cause of the World's Inclination.**

Diogenes and Anaxagoras state that, after the world was composed and had produced living creatures, the world out of its own propensity made an inclination toward the south. Perhaps this may be attributed to a wise Providence (they affirm), that thereby some parts of the world may be habitable, others uninhabitable, according as the various climates are affected with a rigorous cold, or a scorching heat, or a just temperament of cold and heat. Empedocles, that the air yielding to the impetuous force of the solar rays, the poles received an inclination; whereby the northern parts were exalted and the southern depressed, by which means the whole world received its inclination.

## **Chapter IX.**

## **Of that Thing which is Beyond the World, and Whether it be a Vacuum or Not.**

Pythagoras and his followers say that beyond the world there is a vacuum, into which and out of which the world hath its respiration. The Stoics, that there is a vacuum into which infinite space by a conflagration shall be dissolved. Posidonius, not an infinite vacuum, but as much as suffices for the dissolution of the world; and this he asserts in his first book concerning the Vacuum. Aristotle affirms, that a vacuum does not exist. Plato concludes that neither within nor without the world there is any vacuum.

## **Chapter X.**

### **What Parts of the World are on the Right Hand, and what on the Left.**

Pythagoras, Plato, and Aristotle declare that the eastern parts of the world, from whence motion commences, are of the right, those of the western are of the left hand of the world. Empedocles, that those that are of the right hand face the summer solstice, those of the left the winter solstice.

## **Chapter XI.**

### **Of Heaven, what is its Nature and Essence.**

Anaximenes affirms that the circumference of heaven makes the limit of the earth's revolution. Empedocles, that the heaven is a solid substance, and hath the form and hardness of crystal, it being composed of the air compacted by fire, and that in both hemispheres it invests the elements of air and fire. Aristotle, that it is formed by the fifth body, and by the mixture of extreme heat and cold.

## **Chapter XII**

### **Into How Many Circles is the Heaven Distinguished; Or, of the Division of Heaven.**

Thales, Pythagoras, and the followers of Pythagoras do distribute the universal globe of heaven into five circles, which they denominate zones; one of which is called the arctic circle, which is always conspicuous to us, another is the summer tropic, another is the solstice, another is the winter tropic, another is the antarctic circle, which is always out of sight. The circle called the zodiac is placed under the three that are in the midst, and is oblique, gently touching them all. Likewise, they are all divided in right angles by the meridian, which goes from pole to pole. It is supposed that Pythagoras made the first discovery of the obliquity of the zodiac, but one Oenopides of Chios challenges to himself the invention of it.

## **Chapter XIII.**

### **What is the Essence of the Stars, and How They are Composed.**

Thales affirms that they are globes of earth set on fire. Empedocles, that they are fiery bodies arising from that fire which the aether embraced within itself, and did shatter in pieces when the elements were first separated one from another. Anaxagoras, that the circumambient aether is of a fiery substance, which, by a vehement force in its whirling about, did tear stones from the earth, and by its own power set them on fire, and establish them as stars in the heavens. Diogenes thinks they resemble pumice stones, and that they are the breathings of the world; again he supposeth that there are some invisible stones, which fall sometimes from heaven upon the earth, and are there quenched; as it happened at Aegospotami, where a stony star resembling fire did fall. Empedocles, that the fixed stars fastened to the crystal, but the planets are loosened. Plato, that the stars for the most part are of a fiery nature, but they are made partakers of another element, with they are mixed after the resemblance of glue. Zenophanes, that they are composed of inflamed clouds, which in the daytime are quenched, and in the night are kindled again. The like we see in coals; for the rising and setting of the stars is nothing else but the quenching and kindling of them. Heraclitus and the Pythagoreans, that every star is a world in an infinite aether, and encompasseth air, earth, and aether; this opinion is current among the disciples of Orpheus, for they suppose that each of the stars does make a world. Epicurus condemns none of these opinions, for he embraces anything that is possible.

## **Chapter XIV.**

### **Of what Figure the Stars are.**

The Stoics say that the stars are of a circular form, like as the sun, the moon, and the world. Cleanthes, that they are of a conical figure. Anaximenes, that they are fastened as nails in the crystalline firmament; some others, that they are fiery plates of gold, resembling pictures.

## **Chapter XV.**

### **Of the Order and Place of the Stars.**

Xenocrates says that the stars are moved in one and the same superficies. The other Stoics say that they are moved in various superficies, some being superior, others inferior. Democritus, that the fixed stars are in the highest place; after those the planets; after these the sun, Venus, and the moon, in order. Plato, that the first after the fixed stars that makes its appearance is Phaenon, the star of Saturn; the second Phaeton, the star of



Jupiter; the third the fiery, which is the star of Mars; the fourth the morning star, which is the star of Venus; the fifth the shining star, and that is the star of Mercury; in the sixth place is the sun, in the seventh the moon. Plato and some of the mathematicians conspire in the same opinion; others place the sun as the centre of the planets. Anaximander, Metrodorus of Chios, and Crates assign to the sun the superior place, after him the moon, after them the fixed stars and planets.

## **Chapter XVI.**

### **Of the Motion and Circulation of the Stars.**

Anaxagoras, Democritus, and Cleanthes say that all the stars have their motion from east to west. Alcmaeon and the mathematicians, that the planets have a contrary motion to the fixed stars, and in opposition to them are carried from the west to the east. Anaximander, that they are carried by those circles and spheres on which they are placed. Anaximenes, that they are turned under and about the earth. Plato and the mathematicians, that the sun, Venus, and Mercury hold equal measures in their motions.

## **Chapter XVII.**

### **Whence do the Stars Receive Their Light?**

Metrodorus says that all the fixed stars derive their light from the sun. Heraclitus and the Stoics, that earthly exhalations are those by which the stars are nourished. Aristotle, that the heavenly bodies require no nutriment, for they being eternal cannot be obnoxious to corruption. Plato and the Stoics, that the whole world and the stars are fed by the same things.

## **Chapter XVIII.**

### **What are Those Stars which are Called the Dioscuri, the Twins, or Castor and Pollux?**

Xenophanes says that those which appear as stars in the tops of ships are little clouds brilliant by their peculiar motion. Metrodorus, that the eyes of frightened and astonished people emit those lights which are called the Twins.

## **Chapter XIX.**

### **How Stars Prognosticate, and what is the Cause of Winter and Summer.**

Plato says that the summer and winter indications proceed from the rising and setting of the stars, that is, from the rising and setting of the sun, the moon, and the fixed stars. Anaximenes, that the rest in this are not at all concerned, but that it is wholly performed by the sun. Eudoxus and Aratus assign it in common to all the stars, for thus Aratus says:

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Thund'ring Jove stars in heaven hath fixed,  
And them in such beauteous order mixed,  
Which yearly future things predict.

## **Chapter XX.**

### **Of the Essence of the Sun.**

Anaximander says, that the sun is a circle eight and twenty times bigger than the earth, and has a circumference very much like that of a chariot-wheel, which is hollow and full of fire; the fire of which appears to us through its mouth, as by an aperture in a pipe; and this is the sun. Xenophanes, that the sun is constituted of small bodies of fire compacted together and raised from a moist exhalation, which condensed make the body of the sun; or that it is a cloud enfired. The Stoics, that it is an intelligent flame proceeding from the sea. Plato, that it is composed of abundance of fire. Anaxagoras, Democritus, and Metrodorus, that it is an enfired stone, or a burning body. Aristotle, that it is a sphere formed out of the fifth body. Philolaus the Pythagorean, that the sun shines as crystal, which receives its splendor from the fire of the world and so reflecteth its light upon us; so that first, the body of fire which is celestial is in the sun; and secondly, the fiery reflection that comes from it, in the form of a mirror; and lastly, the rays spread upon us by way of reflection from that mirror; and this last we call the sun, which is (as it were) an image of an image. Empedocles, that there are two suns; the one the prototype, which is a fire placed in the other hemisphere, which it totally fills, and is always ordered in a direct opposition to the reflection of its own light; and the sun which is visible to us, formed by the reflection of that splendor in the other hemisphere (which is filled with air mixed with heat), the light reflected from the circular sun in the opposite hemisphere falling upon the crystalline sun; and this reflection is borne round with the motion of the fiery sun. To give briefly the full sense, the sun is nothing else but the light and brightness of that fire which encompasseth the earth. Epicurus, that it is an earthy bulk well compacted, with ores like a pumice-stone or a sponge, kindled by fire.

## **Chapter XXI.**

### **Of the Magnitude of the Sun.**

Anaximander says, that the sun itself in greatness is equal to the earth, but that the circle from whence it receives its respiration and in which it is moved is seven and twenty times larger than the earth. Anaxagoras, that it is far greater than Peloponnesus. Heraclitus, that it is no broader than a man's foot. Epicurus, that he equally embraceth all the foresaid opinions, — that the sun may be of magnitude as it appears, or it may be somewhat greater or somewhat less.

## **Chapter XXII.**

### **What is the Figure or Shape of the Sun.**

Anaximenes affirms that in its dilatation it resembles a leaf. Heraclitus, that it hath the shape of a boat, and is somewhat crooked. The Stoics, that it is spherical, and it is of the same figure with the world and the stars. Epicurus, that the recited dogmas may be defended.

## **Chapter XXIII.**

### **Of the Turning and Returning of the Stars, or the Summer and Winter Solstice.**

Anaximenes believes that the stars are forced by a condensed and resisting air. Anaxagoras, by the repelling force of the northern air, which is violently pushed on by the sun, and thus rendered more condensed and powerful. Empedocles, that the sun is hindered from a continual direct course by its spherical vehicle and by the two circular tropics. Diogenes, that the sun, when it comes to its utmost declination, is extinguished, a rigorous cold damping the heat. The Stoics, that the sun maintains its course only through that space in which its sustenance is seated, let it be the ocean or the earth; by the exhalations proceeding from these it is nourished. Plato, Pythagoras, and Aristotle, that the sun receives a transverse motion from the obliquity of the zodiac, which is guarded by the tropics; all these the globe clearly manifests.

## **Chapter XXIV.**

### **Of the Eclipse of the Sun.**

Thales was the first who affirmed that the eclipse of the sun was caused by the moon's running in a perpendicular line between it and the world; for the moon in its own nature is terrestrial. And by mirrors it is made perspicuous that, when the sun is eclipsed, the moon is in a direct line below it. Anaximander, that the sun is eclipsed when the fiery mouth of it is stopped and hindered from respiration. Heraclitus, that it is after the manner of the turning of a boat, when the concave seems uppermost to our sight, and the convex nethermost. Xenophanes, that the sun is eclipsed when it is extinguished; and

that a new sun is created and rises in the east. He gives a farther account of an eclipse of the sun which remained for a whole month, and again of an eclipse which changed the day into night. Some declare that the cause of an eclipse is the invisible concourse of condensed clouds which cover the orb of the sun. Aristarchus placeth the sun amongst the fixed stars, and believeth that the earth [the moon?] is moved about the sun, and that by its inclination and vergency it intercepts its light and shadows its orb. Xenophanes, that there are many suns and many moons, according as the earth is distinguished by climates, circles, and zones. At some certain times the orb of the sun, falling upon some part of the world which is untenanted, wanders in a vacuum and becomes eclipsed. The same person affirms that the sun proceeding in its motion in the infinite space, appears to us to move orbicularly, taking that representation from its infinite distance from us.

## **Chapter XXV.**

### **Of the Essence of the Moon.**

Anaximander affirms that the circle of the moon is nineteen times bigger than the earth, and resembles the sun, its orb being full of fire; and it suffers an eclipse when the wheel makes a revolution, — which he describes by the divers turnings of a chariot-wheel, in the midst of it there being a hollow nave replenished with fire, which hath but one way of expiration. Xenophanes, that it is a condensed cloud. The Stoics, that it is mixed of fire and air. Plato, that it is a body of the greatest part fiery. Anaxagoras and Democritus, that it is a solid, condensed, and fiery body, in which there are flat countries, mountains, and valleys. Heraclitus, that it is an earth covered with a bright cloud. Pythagoras, that the body of the moon was of a nature resembling a mirror.

## **Chapter XXVI.**

### **Of the Size of the Moon.**

The Stoics declare, that in magnitude it exceeds the earth, just as the sun itself doth. Parmenides, that it is equal to the sun, from whom it receives its light.

## **Chapter XXVII.**

### **Of the Figure of the Moon.**

The Stoics believe that it is of the same figure with the sun, spherical. Empedocles, that the figure of it resembles a quoit. Heraclitus, a boat. Others, a cylinder.

## **Chapter XXVIII.**

## **From Whence is it that the Moon Receives Her Light?**

Anaximander thinks that she gives light to herself, but it is very slender and faint. Antiphon, that the moon shines by its own proper light; but when it absconds itself, the solar beams darting on it obscure it. Thus it naturally happens, that a more vehement light puts out a weaker; the same is seen in other stars. Thales and his followers, that the moon borrows all her light of the sun. Heraclitus, that the sun and moon are after the same manner affected; in their configurations both are shaped like boats, and are made conspicuous to us by receiving their light from moist exhalations. The sun appears to us more refulgent, by reason it is moved in a clearer and purer air; the moon appears more duskish, it being carried in an air more troubled and gross.

### **Chapter XXIX.**

#### **Of the Eclipse of the Moon.**

Anaximenes believes that the mouth of the wheel, about which the moon is turned, being stopped is the cause of an eclipse. Berasus, that it proceeds from the turning of the dark side of the lunar orb towards us. Heraclitus, that it is performed just after the manner of a boat turned upside downwards. Some of the Pythagoreans say, that the splendor arises from the earth, its obstruction from the Antichthon (or counter-earth). Some of the later philosophers, that there is such a distribution of the lunar flame, that it gradually and in a just order burns until it be full moon; in like manner, that this fire decays by degrees, until its conjunction with the sun totally extinguisheth it. Plato, Aristotle, the Stoics, and all the mathematicians agree, that the obscurity with which the moon is every month affected ariseth from a conjunction with the sun, by whose more resplendent beams she is darkened; and the moon is then eclipsed when she falls upon the shadow of the earth, the earth interposing between the sun and moon, or (to speak more properly) the earth intercepting the light of the moon.

### **Chapter XXX.**

#### **Of the Phases of the Moon, or the Lunar Aspects; or How it Comes to Pass that the Moon Appears to Us Terrestrial.**

The Pythagoreans say, that the moon appears to us terraneous, by reason it is inhabited as our earth is, and in it there are animals of a larger size and plants of a rarer beauty than our globe affords; that the animals in their virtue and energy are fifteen degrees superior to ours; that they emit nothing excrementitious; and that the days are fifteen times longer. Anaxagoras, that the reason of the inequality ariseth from the commixture of things earthy and cold; and that fiery and caliginous matter is jumbled together,

whereby the moon is said to be a star of a counterfeit aspect. The Stoics, that on account of the diversity of her substance the composition of her body is subject to corruption.

### **Chapter XXXI.**

#### **How Far the Moon is Removed from the Sun.**

Empedocles declares, that the distance of the moon from the sun is double her remoteness from the earth. The mathematicians, that her distance from the sun is eighteen times her distance from the earth. Eratosthenes, that the sun is remote from the earth seven hundred and eighteen thousand furlongs.

### **Chapter XXXII.**

#### **Of the Year, and How Many Circulations Make Up the Great Year of Every Planet.**

The year of Saturn is completed when he has had his circulation in the space of thirty solar years; of Jupiter in twelve; of Mars in two, of the sun in twelve months; in so many Mercury and Venus, the spaces of their circulation being equal; of the moon in thirty days, in which time her course from her prime to her conjunction is finished. As to the great year, some make it to consist of eight years solar, some of nineteen, others of fifty-nine. Heraclitus, of eighteen thousand. Diogenes, of three hundred and sixty-five such years as Heraclitus assigns. Others there are who lengthen it to seven thousand seven hundred and seventy-seven years.

## **Book III.**

In my two precedent treatises having in due order taken a compendious view and given an account of the celestial bodies, and of the moon which stands between them and the terrestrial, I must now convert my pen to discourse in this third book of Meteors, which are beings above the earth and below the moon, and are extended to the site and situation of the earth, which is supposed to be the centre of the sphere of this world; and from thence will I take my beginning.

### **Chapter I.**

#### **Of the Galaxy, or the Milky Way.**

It is a cloudy circle, which continually appears in the air, and by reason of the whiteness of its colors is called the galaxy, or the milky way. Some of the Pythagoreans say that, when Phaeton set the world on fire, a star falling from its own place in its circular passage through the region caused an inflammation. Others say that originally it was the first course of the sun; others, that it is an image as in a looking-glass, occasioned by the sun's reflecting its beams towards the heavens, and this appears in the clouds and in the rainbow. Metrodorus, that it is merely the solar course, or the motion of the sun in its own circle. Parmenides, that the mixture of a thick and thin substance gives it a color which resembles milk. Anaxagoras, that the sun moving under the earth and not being able to enlighten every place, the shadow of the earth, being cast upon the part of the heavens, makes the galaxy. Democritus, that it is the splendor which ariseth from the coalition of many small bodies, which, being firmly united amongst themselves, do mutually enlighten one another. Aristotle, that it is the inflammation of dry, copious, and coherent vapor, by which the fiery mane, whose seat is beneath the aether and the planets, is produced. Posidonius, that it is a combination of fire, of finer substance than the stars, but denser than light.

### **Chapter II.**

#### **Of Comets and Shooting Fires, and Those which Resemble Beams.**

Some of the Pythagoreans say, that a comet is one of those stars which do not always appear, but after they have run through their determined course, they then rise and are visible to us. Others, that it is the reflection of our sight upon the sun, which gives the resemblance of comets much after the same manner as images are reflected in mirrors. Anaxagoras and Democritus, that two or more stars being in conjunction by their united light make a comet. Aristotle, that it is a fiery coalition of dry exhalations. Strato, that it is the light of the star darting through a thick cloud that hath invested it; this is seen in

light shining through lanterns. Heraclides, native of Pontus, that it is a lofty cloud inflamed by a sublime fire. The like causes he assigns to the bearded comet, to those circles that are seen about the sun or stars, or those meteors which resemble pillars or beams, and all others which are of this kind. This way unanimously go all the Peripatetics, holding that these meteors, being formed by the clouds, do differ according to their various configurations. Epigenes, that a comet arises from a rising of spirit or wind, mixed with an earthy substance and set on fire. Boethus, that it is a phantasy presented to us by fiery air. Diogenes, that comets are stars. Anaxagoras, that those styled shooting stars descend from the aether like sparks, and therefore are soon extinguished. Metrodorus, that it is a forcible illapse of the sun upon clouds which makes them to sparkle as fire. Xenophanes, that all such fiery meteors are nothing else but the conglomeration of the enfi red clouds, and the flashing motions of them.

### **Chapter III.**

#### **Of Violent Eruption of Fire Out of the Clouds. of Lightning. of Thunder. of Hurricanes. of Whirlwinds.**

Anaximander affirms that all these are produced by the wind after this manner: the wind being enclosed by condensed clouds, on account of its minuteness and lightness violently endeavors to make a passage; and in breaking through the cloud gives noise; and the tearing the cloud, because of the blackness of it, gives a resplendent flame. Metrodorus, that when the wind falls upon a cloud whose densing firmly compacts it, by breaking the cloud it causeth a great noise, and by striking and rending the cloud it gives the flame; and in the swiftness of its motion, the sun imparting heat to it, it throws out the bolt. The weak declining of the thunderbolt ends in a violent tempest. Anaxagoras, that when heat and cold meet and are mixed together (that is, ethereal parts with airy), thereby a great noise of thunder is produced, and the color observed against the blackness of the cloud occasions the flashing of fire; the full and great splendor is lightning, the more enlarged and embodied fire becomes a whirlwind, the cloudiness of it gives the hurricane. The Stoics, that thunder is the clashing of clouds one upon another, the flash of lightning is their fiery inflammation; their more rapid splendor is the thunderbolt, the faint and weak the whirlwind. Aristotle, that all these proceed from dry exhalations, which, if they meet with moist vapors, forcing their passage, the breaking of them gives the noise of thunder; they, being very dry, take fire and make lightning; tempests and hurricanes arise from the plenitude of matter which each draw to themselves, the hotter parts attracted make the whirlwinds, the duller the tempests.

### **Chapter IV.**

#### **Of Clouds, Rain, Snow, and Hail.**



Anaximenes thinks that the air by being very much condensed clouds are formed; this air being more compacted, rain is compressed through it; when water in its falling down freezeth, then snow is generated; when it is encompassed with a moist air, it is hail. Metrodorus, that a cloud is composed of a watery exhalation carried into a higher place. Epicurus, that they are made of vapors; and that hail and snow are formed in a round figure, being in their long descent pressed upon by the circumambient air.

## Chapter V.

### Of the Rainbow.

Those things which affect the air in the superior places of it are of two sorts. Some have a real subsistence, such are rain and hail; others not. Those which enjoy not a proper subsistence are only in appearance; of this sort is the rainbow. Thus the continent to us that sail seems to be in motion.

Plato says, that men admiring it feigned that it took origination from one Thaumás, which word signifies admiration. Homer sings: —

Jove paints the rainbow with a purple dye,  
Alluring man to cast his wandering eye.  
(*Iliad*, xvii. 547.)

Others therefore fabled that the bow hath a head like a bull, by which it swallows up rivers.

But what is the cause of the rainbow? It is evident that what apparent things we see come to our eyes in right or in crooked lines, or by refraction: these are incorporeal and to sense obscure, but to reason they are obvious. Those which are seen in right lines are those which we see through the air or horn or transparent stones, for all the parts of these things are very fine and tenuous; but those which appear in crooked lines are in water, the thickness of the water presenting them bended to our sight. This is the reason that oars in themselves straight, when put into the sea, appear to us crooked. The third manner of our seeing is by refraction, and this is perspicuous in mirrors. After this third sort the rainbow is affected. We conceive it is a moist exhalation converted into a cloud, and in a short space it is dissolved into small and moist drops. The sun declining towards the west, it will necessarily follow that the whole bow is seen opposite to the sun; for the eye being directed to those drops receives a refraction, and by this means the bow is formed. The eye doth not consider the figure and form, but the color of these drops; the first of which colors is a shining red, the second a purple, the third is blue and green. Let us consider whether the reason of this red shining color be the splendor of the sun falling upon these small drops, the whole body of light being refracted, by which this bright red color is produced; the second part being troubled and the light

languishing in the drops, the color becomes purple (for the purple is the faint red); but the third part, being more and more troubled, is changed into the green color. And this is proved by other effects of Nature; if any one shall put water in his mouth and spit it out so opposite to the sun, that its rays may be refracted on the drops, he shall see the resemblance of a rainbow; the same appears to men that are blear-eyed, when they fix their watery eyes upon a candle. Anaximenes thinks the bow is thus formed; the sun casting its splendor upon a thick, black, and gross cloud, and the rays not being in a capacity to penetrate beyond the superficies. Anaxagoras, that, the solar rays being reflected from a condensed cloud, the sun being placed directly opposite to it forms the bow after the mode of the repercussion of a mirror; after the same manner he assigns the natural cause of the Parhelia or mock-suns, which are often seen in Pontus. Metrodorus, that when the sun casts its splendor through a cloud, the cloud gives itself a blue, and the light a red color.

## **Chapter VI.**

### **Of Meteors which Resemble Rods, or of Rods.**

These rods and the mock-suns are constituted of a double nature, a real subsistence, and a mere appearance; — of a real subsistence, because the clouds are the object of our eyes; of a mere appearance, for their proper color is not seen, but that which is adventitious. The like affections, natural and adventitious, in all such things do happen.

## **Chapter VII.**

### **Of Winds.**

Anaximander believes that wind is a fluid air, the sun putting into motion or melting the moist subtle parts of it. The Stoics, that all winds are a flowing air, and from the diversity of the regions whence they have their origin receive their denomination; as, from darkness and the west the western wind; from the sun and its rising the eastern; from the north the northern, and from the south the southern winds. Metrodorus, that moist vapors heated by the sun are the cause of the impetuosity of violent winds. The Etesian, or those winds which annually commence about the rising of the Little Dog, the air about the northern pole being more compacted, blow violently following the sun when it returns from the summer solstice.

## **Chapter VIII.**

### **Of Winter and Summer.**

Empedocles and the Stoics believe that winter is caused by the thickness of the air prevailing and mounting upwards; and summer by fire, it falling downwards.

This description being given by me of Meteors, or those things that are above us, I must pass to those things which are terrestrial.

## **Chapter IX.**

### **Of the Earth, what is its Nature and Magnitude.**

Thales and his followers say that there is but one earth. Hicetes the Pythagorean, that there are two earths, this and the Antichthon, or the earth opposite to it. The Stoics, that this earth is one, and that finite and limited. Xenophanes, that the earth, being compacted of fire and air, in its lowest parts hath laid a foundation in an infinite depth. Metrodorus, that the earth is mere sediment and dregs of water, as the sun is of the air.

## **Chapter X.**

### **Of the Figure of the Earth.**

Thales, the Stoics, and their followers say that the earth is globular. Anaximander, that it resembles a smooth stony pillar. Anaximenes, that it hath the shape of a table. Leucippus, of a drum. Democritus, that it is like a quoit externally, and hollow in the middle.

## **Chapter XI.**

### **Of the Site and Position of the Earth.**

The disciples of Thales say that the earth is the centre of the universe. Xenophanes, that it is first, being rooted in the infinite space. Philolaus the Pythagorean gives to fire the middle place, and this is the source fire of the universe; the second place to the Antichthon; the third to that earth which we inhabit, which is placed in opposition unto and whirled about the opposite, — which is the reason that those which inhabit that earth cannot be seen by us. Parmenides was the first that confined the habitable world to the two solstitial (or temperate) zones.

## **Chapter XII.**

### **Of the Inclination of the Earth.**

Leucippus affirms that the earth vergeth towards the southern parts, by reason of the thinness and fineness that is in the south; the northern parts are more compacted, they

being congealed by a rigorous cold, but those parts of the world that are opposite are enfi red. Democritus, because, the southern parts of the air being the weaker, the earth as it enlarges bends towards the south; the northern parts are of an unequal, the southern of an equal temperament; and this is the reason that the earth bends towards those parts where the earth is laden with fruits and its own increase.

### **Chapter XIII.**

#### **Of the Motion of the Earth.**

Most of the philosophers say that the earth remains fixed in the same place. Philolaus the Pythagorean, that it is moved about the element of fire, in an oblique circle, after the same manner of motion that the sun and moon have. Heraclides of Pontus and Ecphantus the Pythagorean assign a motion to the earth, but not progressive, but after the manner of a wheel being carried on its own axis; thus the earth (they say) turns itself upon its own centre from west to east. Democritus, that when the earth was first formed it had a motion, the parts of it being small and light; but in process of time the parts of it were condensed, so that by its own weight it was poised and fixed.

### **Chapter XIV.**

#### **Into How Many Zones is the Earth Divided?**

Pythagoras says that, as the celestial sphere is distributed into five zones, into the same number is the terrestrial; which zones are the arctic and antarctic, the summer and winter tropics (or temperate zones), and the equinoctial; the middle of which zones equally divides the earth and constitutes the torrid zone; but that portion which is in between the summer and winter tropics is habitable, by reason the air is there temperate.

### **Chapter XV.**

#### **Of Earthquakes.**

Thales and Democritus assign the cause of earthquakes to water. The Stoics say that it is a moist vapor contained in the earth, making an irruption into the air, that causes the earthquake. Anaximenes, that the dryness and rarity of the earth are the cause of earthquakes, the one of which is produced by extreme drought, the other by immoderate showers. Anaxagoras, that the air endeavoring to make a passage out of the earth, meeting with a thick superficies, is not able to force its way, and so shakes the circumambient earth with a trembling. Aristotle, that a cold vapor encompassing every part of the earth prohibits the evacuation of vapors; for those which are hot, being in

themselves light, endeavor to force a passage upwards, by which means the dry exhalations, being left in the earth, use their utmost endeavor to make a passage out, and being wedged in, they suffer various circumvolutions and shake the earth. Metrodorus, that whatsoever is in its own place is incapable of motion, except it be pressed upon or drawn by the operation of another body; the earth being so seated cannot naturally be moved, yet divers parts and places of the earth may move one upon another. Parmenides and Democritus, that the earth being so equally poised hath no sufficient ground why it should incline more to one side than to the other; so that it may be shaken, but cannot be removed. Anaximenes, that the earth by reason of its latitude is borne upon by the air which presseth upon it. Others opine that the earth swims upon the waters, as boards and broad planks, and by that reason is moved. Plato, that motion is by six manner of ways, upwards, downwards, on the right hand and on the left, behind and before; therefore it is not possible that the earth should be moved in any of these modes, for it is altogether seated in the lowest place; it therefore cannot receive a motion, since there is nothing about it so peculiar as to cause it to incline any way; but some parts of it are so rare and thin that they are capable of motion. Epicurus, that the possibility of the earth's motion ariseth from a thick and aqueous air under the earth, that may, by moving or pushing it, be capable of quaking; or that being so compassed, and having many passages, it is shaken by the wind which is dispersed through the hollow dens of it.

## **Chapter XVI.**

### **Of the Sea, and How it is Composed, and How it Becomes to the Taste Bitter.**

Anaximander affirms that the sea is the remainder of the primogenial humidity, the greatest part of which being dried up by the fire, the influence of the great heat altered its quality. Anaxagoras that in the beginning water did not flow, but was as a standing pool; and that it was burnt by the movement of the sun about it, by which the oily part of the water being exhaled, the residue became salt. Empedocles, that the sea is the sweat of the earth heated by the sun. Antiphon, that the sweat of that which was hot was separated from the rest which were moist; these by seething and boiling became bitter, as happens in all sweats. Metrodorus, that the sea was strained through the earth, and retained some part of its density; the same is observed in all those things which are strained through ashes. The schools of Plato, that the element of water being compacted by the rigor of the air became sweet, but that part which was expired from the earth, being enfired, became of a brackish taste.

## **Chapter XVII.**

### **Of Tides, or of the Ebbing and Flowing of the Sea.**

Aristotle and Heraclides say, they proceed from the sun, which moves and whirls about the winds; and these falling with a violence upon the Atlantic, it is pressed and swells by them, by which means the sea flows; and their impression ceasing, the sea retracts, hence they ebb. Pytheas the Massilian, that the fulness of the moon gives the flow, the wane the ebb. Plato attributes it all to a certain balance of the sea, which by means of a mouth or orifice causes the tide; and by this means the seas do rise and flow alternately. Timaeus believes that those rivers which fall from the mountains of the Celtic Gaul into the Atlantic produce a tide. For upon their entering upon that sea, they violently press upon it, and so cause the flow; but they disembodying themselves, there is a cessation of the impetuosity, by which means the ebb is produced. Seleucus the mathematician attributes a motion to the earth; and thus he pronounceth that the moon in its circulation meets and repels the earth in its motion; between these two, the earth and the moon, there is a vehement wind raised and intercepted, which rushes upon the Atlantic Ocean, and gives us a probable argument that it is the cause the sea is troubled and moved.

## **Chapter XVIII.**

### **Of the Aurea, or a Circle About a Star.**

The aurea or circle is thus formed. A thick and dark air intervening between the moon or any other star and our eye, by which means our sight is dilated and reflected, when now our sight falls upon the outward circumference of the orb of that star, there presently seems a circle to appear. This circle thus appearing is called the [Greek omitted] or halo; and there is constantly such a circle seen by us, when such a density of sight happens.

## **Book IV.**

Having taken a survey of the general parts of the world, I will take a view of the particular members of it.

### **Chapter I.**

#### **Of the Overflowing of the Nile.**

Thales conjectures that the Etesian or anniversary northern winds blowing strongly against Egypt heighten the swelling of the Nile, the mouth of that river being obstructed by the force of the sea rushing into it. Euthymenes the Massilian concludes that the Nile is filled by the ocean and that sea which is outward from it, the last being naturally sweet. Anaxagoras, that the snow in Ethiopia which is frozen in winter is melted in summer, and this makes the inundation. Democritus, that the snows which are in the northern climates when the sun enters the summer solstice are dissolved and diffused; from those vapors clouds are compacted, and these are forcibly driven by the Etesian winds into the southern parts and into Egypt, from whence violent showers are poured; and by this means the fens of Egypt are filled with water, and the river Nile hath its inundation. Herodotus the historian, that the waters of the Nile receive from their fountain an equal portion of water in winter and in summer; but in winter the water appears less, because the sun, making its approach nearer to Egypt, draws up the rivers of that country into exhalation. Ephorus the historiographer, that in summer all Egypt seems to be melted and sweats itself into water, to which the thin and sandy soils of Arabia and Lybia contribute. Eudoxus relates that the Egyptian priests affirm that, when it is summer to us who dwell under the northern tropic, it is winter with them that inhabit under the southern tropic; by this means there is a various contrariety and opposition of the seasons in the year, which cause such showers to fall as make the water to overflow the banks of the Nile and diffuse itself throughout all Egypt.

### **Chapter II.**

#### **Of the Soul.**

Thales first pronounced that the soul is that being which is in a perpetual motion, or that whose motion proceeds from itself. Pythagoras, that it is a number moving itself; he takes a number to be the same thing with a mind. Plato, that it is an intellectual substance moving itself, and that motion is in a numerical harmony. Aristotle, that it is the first actuality [Greek omitted] of a natural organical body which has life potentially; and this actuality must be understood to be the same thing with energy or

operation. Dicaearchus, that it is the harmony of the four elements. Asclepiades the physician, that it is the concurrent exercitation of the senses.

### **Chapter III.**

#### **Whether the Soul be a Body, and what is the Nature and Essence of It.**

All those named by me do affirm that the soul itself is incorporeal, and by its own nature is in a motion, and in its own self is an intelligent substance, and the living actuality of a natural organical body. The followers of Anaxagoras, that it is airy and a body. The Stoics, that it is a hot exhalation. Democritus, that it is a fiery composition of things which are perceptible by reason alone, the same having their forms spherical and without an inflaming faculty; and it is a body. Epicurus, that it is constituted of four qualities, of a fiery quality, of an aerial quality, a pneumatical, and of a fourth quality which hath no name, but it contains the virtue of the sense. Heraclitus, that the soul of the world is the exhalation which proceeds from the moist parts of it; but the soul of animals, arising from exhalations that are exterior and from those that are within them, is homogeneous to it.

### **Chapter IV.**

#### **Of the Parts of the Soul.**

Plato and Pythagoras, according to their first account, distribute the soul into two parts, the rational and irrational. By a more accurate and strict account the soul is branched into three parts; they divide the unreasonable part into the concupiscible and the irascible. The Stoics say the soul is constituted of eight parts; five of which are the senses, hearing, seeing, tasting, touching, smelling, the sixth is the faculty of speaking, the seventh of generating, the eighth of commanding; this is the principal of all, by which all the other are guided and ordered in their proper organs, as we see the eight arms of a polypus aptly disposed. Democritus and Epicurus divide the soul into two parts, the one rational, which hath its residence in the breast, and the irrational, which is diffused through the whole structure of the body. Democritus, that the quality of the soul is communicated to everything, yea, to the dead corpses; for they are partakers of heat and some sense, when the most of both is expired out of them.

### **Chapter V.**

#### **What is the Principal Part of the Soul, and in what Part of the Body it Resides.**

Plato and Democritus place its residence in the whole head. Strato, in that part of the forehead where the eyebrows are separated. Erasistratus, in the Epikranis, or membrane



which involves the brain. Herophilus, in that sinus of the brain which is the basis of it. Parmenides, in the breast; which opinion is embraced by Epicurus. The Stoics are generally of this opinion, that the seat of the soul is throughout the heart, or in the spirit about it. Diogenes, in the arterial ventricle of the heart, which is also full of vital spirit. Empedocles, in the mass of the blood. There are that say it is in the neck of the heart, others in the pericardium, others in the midriff. Certain of the Neoterics, that the seat of the soul is extended from the head to the diaphragm. Pythagoras, that the animal part of the soul resides in the heart, the intellectual in the head.

## **Chapter VI.**

### **Of the Motion of the Soul.**

Plato believes that the soul is in perpetual motion, but that it is immovable as regards motion from place to place. Aristotle, that the soul is not naturally moved, but its motion is accidental, resembling that which is in the forms of bodies.

## **Chapter VII.**

### **Of the Soul's Immortality.**

Plato and Pythagoras say that the soul is immortal; when it departs out of the body, it retreats to the soul of the world, which is a being of the same nature with it. The Stoics, when the souls leave the bodies, they are carried to divers places; the souls of the unlearned and ignorant descend to the coagmentation of earthly things, but the learned and vigorous last till the general fire. Epicurus and Democritus, the soul is mortal, and it perisheth with the body. Plato and Pythagoras, that part of the soul of man which is rational is eternal; for though it be not God, yet it is the product of an eternal deity; but that part of the soul which is divested of reason dies.

## **Chapter VIII.**

### **Of the Senses, and of Those Things which are Objects of the Senses,**

The Stoics give this definition of sense: Sense is the Apprehension or comprehension of an object by means of an organ of sensation. There are several ways of expressing what sense is; it is either a habit, a faculty, an operation, or an imagination which apprehends by means of an organ of sense, — and also the eighth principal thing, from whence the senses originate. The instruments of sense are intelligent exhalations, which from the said commanding part extend unto all the organs of the body. Epicurus, that sense is a faculty, and that which is perceived by the sense is the product of it; so that sense hath a double acceptance, — sense which is the faculty, and the thing received by the sense,

which is the effect. Plato, that sense is that commerce which the soul and body have with those things that are exterior to them; the power of which is from the soul, the organ by which is from the body; but both of them apprehend external objects by means of the imagination. Leucippus and Democritus, that sense and intelligence arise from external images; so neither of them can operate without the assistance of image falling upon us.

## **Chapter IX.**

### **Whether what Appears to Our Senses and Imaginations be True or Not.**

The Stoics say that what the senses represent is true; what the imagination, is partly false, partly true. Epicurus that every impression of the sense or imagination is true, but of those things that fall under the head of opinion, some are true, some false: sense gives us a false presentation of those things only which are the objects of our understanding; but the imagination gives us a double error, both of things sensible and things intellectual. Empedocles and Heraclides, that the senses act by a just accommodation of the pores in every case; everything that is perceived by the sense being congruously adapted to its proper organ.

## **Chapter X.**

### **How Many Senses are There?**

The Stoics say that there are five senses properly so called, seeing, hearing, smelling, tasting. and touching. Aristotle indeed doth not add a sixth sense; but he assigns a common sense, which is the judge of all compounded species; into this each sense casts its proper representation, in which is discovered a transition of one thing into another, like as we see in figure and motion where there is a change of one into another. Democritus, that there are divers species of senses, which appertain to beings destitute of reason, to the gods, and to wise men.

## **Chapter XI.**

### **How the Actions of the Senses, the Conceptions of Our Minds, and the Habit of Our Reason are Formed.**

The Stoics affirm that every man, as soon as he is born, has a principal and commanding part of his soul, which is in him like a sheet of writing-paper, to which he commits all his notions. The first manner of his inscribing is by denoting those notions which flow from the senses. Suppose it be of a thing that is white; when the present sense of it is vanished, there is yet retained the remembrance; when many memorative notions of the

same similitude do concur, then he is said to have an experience; for experience is nothing more than the abundance of notions that are of the same form met together. Some of these notions are naturally begotten according to the aforesaid manner, without the assistance of art; the others are produced by discipline, learning, and industry; these only are justly called notions, the others are prenotions. But reason, which gives us the denomination of rational, is completed by prenotions in the first seven years. The conception of the mind is the vision that the intelligence of a rational animal hath received; when that vision falls upon the rational soul, then it is called the conception of the mind, for it hath derived its name from the mind [Greek omitted] from [Greek omitted]. Therefore these visions are not to be found in any other animals; they only are appropriated to gods and to us men. If these we consider generally, they are phantasms; if specifically, they are notions. As pence or staters, if you consider them according to their own value, are simply pence and staters; but if you give them as a price for a naval voyage, they are called not merely pence, etc., but your freight.

## **Chapter XII.**

### **What is the Difference Between Imagination [Greek Omitted], The Imaginable [GREEK OMITTED], Fancy [GREEK OMITTED], and Phantom [Greek Omitted]?**

Chrysippus affirms, these four are different one from another. Imagination is that passion raised in the soul which discovers itself and that which was the efficient of it; to use example, after the eye hath looked upon a thing that is white, the sight of which produceth in the mind a certain impression, this gives us reason to conclude that the object of this impression is white, which affecteth us. So with touching and smelling Phantasy or imagination is denominated from [Greek omitted] which denotes light; for as light discovers itself and all other things which it illuminates, so this imagination discovers itself and that which is the cause of it. The imaginable is the efficient cause of imagination; as anything that is white, or anything that is cold, or everything that may make an impression upon the imagination. Fancy is a vain impulse upon the mind of man, proceeding from nothing which is really conceivable; this is experienced in those that whirl about their idle hand and fight with shadows; for to the imagination there is always some real imaginable thing presented, which is the efficient cause of it; but to the fancy nothing. A phantom is that to which we are brought by such a fanciful and vain attraction; this is to be seen in melancholy and distracted persons. Of this sort was Orestes in the tragedy, pronouncing these words:

Mother, these maids with horror me affright;  
Oh bring them not, I pray, into my sight!

They're smeared with blood, and cruel, dragon-like,  
Skipping about with deadly fury strike.

These rave as frantic persons, they see nothing, and yet imagine they see. Thence Electra thus returns to him:

o wretched man, securely sleep in bed;  
Nothing thou seest, thy fancy's vainly led.  
(Euripides, "Orestes", 255.)

After the same manner Theoclymenus in Homer.

### **Chapter XIII.**

#### **Of Our Sight, and by what Means We See.**

Democritus and Epicurus suppose that sight is caused by the insertion of little images into the visive organ, and by the reception of certain rays which return to the eye after meeting the object. Empedocles supposes that images are mixed with the rays of the eye; these he styles the rays of images. Hipparchus, that the visual rays extend from both the eyes to the superficies of bodies, and give to the sight the apprehension of those same bodies, after the same manner in which the hand touching the extremity of bodies gives the sense of feeling. Plato, that the sight is the splendor of united rays; there is a light which reaches some distance from the eyes into a cognate air, and there is likewise a light shed from bodies, which meets and joins with the fiery visual light in the intermediate air (which is liquid and mutable); and the union of these rays gives the sense of seeing. This is Plato's corradiancy, or splendor of united rays.

### **Chapter XIV.**

#### **Of Those Images which are Presented to Our Eyes in Mirrors.**

Empedocles says that these images are caused by certain effluxes which, meeting together and resting upon the superficies of the mirror, are perfected by that fiery element emitted by the said mirror, which transforms withal the air that surrounds it. Democritus and Epicurus, that the specular appearances are made by the subsistence of the images which flow from our eyes; these fall upon the mirror and remain, while the light returns to the eye. The followers of Pythagoras explain it by the reflection of the sight; for our sight being extended (as it were) to the brass, and meeting with the smooth dense surface thereof it is forced back, and caused to return upon itself: the same takes place in the hand, when it is stretched out and then brought back again to the shoulder. Any one may use these instances to explain the manner of seeing.

## **Chapter XV.**

### **Whether Darkness Can be Visible to Us.**

The Stoics say that darkness is seen by us, for out of our eyes there issues out some light into it; and our eyes do not impose upon us, for they really perceive there is darkness. Chrysippus says that we see darkness by the striking of the intermediate air; for the visual spirits which proceed from the principal part of the soul and reach to the ball of the eye pierce this air, which, after they have made those strokes upon it, extend conically on the surrounding air, where this is homogeneous in quality. For from the eyes those rays are poured forth which are neither black nor cloudy. Upon this account darkness is visible to us.

## **Chapter XVI.**

### **Of Hearing.**

Empedocles says that hearing is formed by the insidency of the air upon the cochlea, which it is said hangs within the ear as a bell, and is beat upon by the air. Alcmaeon, that the vacuity that is within the ear makes us to have the sense of hearing, for the air forcing a vacuum gives the sound; every inanity affords a ringing. Diogenes the air which exists in the head, being struck upon by the voice gives the hearing. Plato and his followers, the air which exists in the head being struck upon, is reflected to the principal part of the soul, and this causeth the sense of hearing.

## **Chapter XVII.**

### **Of Smelling.**

Alcmaeon believes that the principal part of the soul, residing in the brain, draws to itself odors by respiration. Empedocles, that scents insert themselves into the breathing of the lungs; for, when there is a great difficulty in breathing, odors are not perceived by reason of the sharpness; and this we experience in those who have the defluxion of rheum.

## **Chapter XVIII.**

### **Of Taste.**

Alcmaeon says that a moist warmth in the tongue, joined with the softness of it, gives the difference of taste. Diogenes, that by the softness and sponginess of the tongue, and because the veins of the body are joined in it, tastes are diffused by the tongue; for they

are attracted from it to that sense and to the commanding part of the soul, as from a sponge.

## **Chapter XIX.**

### **Of the Voice.**

Plato thus defines a voice, — that it is a breath drawn by the mind through the mouth, and a blow impressed on the air and through the ear, brain, and blood transmitted to the soul. Voice is abusively attributed to irrational and inanimate beings; thus we improperly call the neighing of horses or any other sound by the name of voice. But properly a voice [Greek omitted] is an articulate sound, which illustrates [Greek omitted] the understanding of man. Epicurus says that it is an efflux emitted from things that are vocal, or that give sounds or great noises; this is broken into those fragments which are after the same configuration. Like figures are round figures with round, and irregular and triangular with those of the same kind. These falling upon the ears produce the sense of hearing. This is seen in leaking vessels, and in fullers when they fan or blow their cloths. Democritus, that the air is broken into bodies of similar configuration, and these are rolled up and down with the fragments of the voice; as it is proverbially said, One daw lights with another, or, God always brings like to like. Thus we see upon the seashore, that stones like to one another are found in the same place, in one place the long-shaped, in another the round are seen. So in sieves, things of the same form meet together, but those that are different are divided; as pulse and beans falling from the same sieve are separated one from another. To this it may be objected: How can some fragments of air fill a theatre in which there is an infinite company of persons. The Stoics, that the air is not composed of small fragments, but is a continued body and nowhere admits a vacuum; and being struck with the air, it is infinitely moved in waves and in right circles, until it fill that air which surrounds it; as we see in a fish-pool which we smite by a falling stone cast upon it; yet the air is moved spherically, the water orbicularly. Anaxagoras says a voice is then formed when upon a solid air the breath is incident, which being repercussed is carried to the ears; after the same manner the echo is produced.

## **Chapter XX.**

### **Whether the Voice is Incorporeal. what is it that the Gives Echo?**

Pythagoras, Plato, and Aristotle declare that the voice is incorporeal; for it is not the air that causes the voice, but the figure which compasseth the air and its superficies having received a stroke, give the voice. But every superficies of itself is incorporeal. It is true that it move with the body but itself it hath no body; as we observe in a staff that is bended, the matter only admits of an inflection, while the superficies doth not.

According to the Stoics a voice is corporeal since everything that is an agent or operates is a body; a voice acts and operates, for we hear it and are sensible of it; for it falls and makes an impression on the ear, as a seal of a ring gives its similitude upon the wax. Besides, everything that creates a delight or injury is a body; harmonious music affects with delight, but discord is tiresome. And everything that moved is a body; and the voice moves, and having its illapse upon smooth places is reflected, as when a ball is cast against a wall it rebounds. A voice spoken in the Egyptian pyramids is so broken, that it gives four or five echoes.

## **Chapter XXI.**

### **By what Means the Soul is Sensible, and what is the Principal and Commanding Part of It.**

The Stoics say that the highest part of the soul is the commanding part of it: this is the cause of sense, fancy, consents, and desires; and this we call the rational part. From this principal and commander there are produced seven parts of the soul, which are spread through the body, as the seven arms in a polypus. Of these seven parts, five are assigned to the senses, seeing, hearing, smelling, tasting, touching. Sight is a spirit which is extended from the commanding part of the eyes; hearing is that spirit which from the principle reacheth to the ears; smelling a spirit drawn from the principal to the nostrils; tasting a spirit extended from the principle to the tongue; touching is a spirit which from the principal is drawn to the extremity of those bodies which are obnoxious to a sensible touch. Of the rest, the one called the spermatical is a spirit which reacheth from the principal to the generating vessels; the other, which is the vocal and termed the voice, is a spirit extended from the principal to the throat, tongue, and other proper organs of speaking. And this principal part itself hath that place in our spherical head which God hath in the world.

## **Chapter XXII.**

### **Of Respiration or Breathing.**

Empedocles thinks, that the first breath the first animal drew was when the moisture in the embryo was separated, and by that means an entrance was given to the external air into the gaping vessels, the moisture in them being evacuated. After this the natural heat, in a violent force pressing upon the external air for a passage, begets an expiration; but this heat returning to the inward parts, and the air giving way to it, causeth a respiration. The respiration that now is arises when the blood is borne to the exterior surface, and by this movement drives the airy substance through the nostrils; thus in its recess it causeth expiration, but the air being again forced into those places which are emptied of

blood, it causeth an inspiration. To explain which, he proposeth the instance of a water-clock, which gives the account of time by the running of water.

Asclepiades supposeth the lungs to be in the manner of a funnel, and the cause of breathing to be the fineness of the inward parts of the breast; for thither the outward air which is more gross hastens, but is forced backward, the breast not being capable either to receive or want it. But there being always some of the more tenuous parts of the air left, so that all of it is not exploded, to that which there remains the more ponderous external air with equal violence is forced; and this he compares to cupping-glasses. All spontaneous breathings are formed by the contracting of the smaller pores of the lungs, and to the closing of the pipe in the neck; for these are at our command.

Herophilus attributes a moving faculty to the nerves, arteries, and muscles, but thinks that the lungs are affected only with a natural desire of enlarging and contracting themselves. Farther, there is the first operation of the lungs by attraction of the outward air, which is drawn in because of the abundance of the external air. Next to this, there is a second natural appetite of the lungs; the breast, pouring in upon itself the breath, and being filled, is no longer able to make an attraction, and throws the superfluity of it upon the lungs, whereby it is then sent forth in expiration; the parts of the body mutually concurring to this function by the alternate participation of fulness and emptiness. So that to lungs pertain four motions — first, when the lungs receive the outward air; secondly, when the outward air thus entertained is transmitted to the breast; thirdly, when the lungs again receive that air which they imparted to the breast; fourthly, when this air then received from the breast is thrown outwards. Of these four processes two are dilatations, one when the lungs attract the air, another when the breast dischargeth itself of it upon the lungs; two are contractions, one when the breast draws into itself the air, the second when it expels this which was insinuated into it. The breast admits only of two motions — of dilatation, when it draws from the lungs the breath, and of contraction, when it returns what it did receive.

## **Chapter XXIII.**

### **Of the Passions of the Body, and Whether the Soul Hath a Sympathetical Condolency with It.**

The Stoics say that all the passions are seated in those parts of the body which are affected, the senses have their residence in the commanding part of the soul. Epicurus, that all the passions and all the senses are in those parts which are affected, but the commanding part is subject to no passion. Strato, that all the passions and senses of the soul are in the rational or commanding part of it, and are not fixed in those places which are affected; for in this place patience takes its residence, and this is apparent in terrible and dolorous things, as also in timorous and valiant individuals.



## **Book V.**

### **Chapter I.**

#### **Of Divination.**

Plato and the Stoics introduce divination as a godlike enthusiasm, the soul itself being of a divine constitution, and this prophetic faculty being inspiration, or an illapse of the divine knowledge into man; and so likewise they account for interpretation by dreams. And these same allow many divisions of the art of divination. Xenophanes and Epicurus utterly refuse any such art of foretelling future contingencies. Pythagoras rejects all manner of divination which is by sacrifices. Aristotle and Dicaearchus admit only these two kinds of it, a fury by a divine inspiration, and dreams; they deny the immortality of the soul, yet they affirm that the mind of man hath a participation of something that is divine.

### **Chapter II.**

#### **Whence Dreams do Arise.**

Democritus says that dreams are formed by the illapse of adventitious representations. Strato, that the irrational part of the soul in sleep becoming more sensible is moved by the rational part of it. Herophilus, that dreams which are caused by divine instinct have a necessary cause; but dreams which have their origin from a natural cause arise from the soul's forming within itself the images of those things which are convenient for it, and which will happen; those dreams which are of a constitution mixed of both these have their origin from the fortuitous appulse of images, as when we see those things which please us; thus it happens many times to those persons who in their sleep imagine they embrace their mistresses.

### **Chapter III.**

#### **Of the Nature of Generative Seed.**

Aristotle says, that seed is that thing which contains in itself a power of moving, whereby it is enabled to produce a being like unto that from whence it was emitted. Pythagoras, that seed is the sediment of that which nourisheth us, the froth of the purest blood, of the same nature of the blood and marrow of our bodies. Alcmaeon, that it is part of the brain. Plato, that it is the deflux of the spinal marrow. Epicurus, that it is a fragment torn from the body and soul. Democritus, that it proceeds from all the parts of the body, and chiefly from the principal parts, as the tissues and muscles.

## **Chapter IV.**

### **Whether the Sperm be a Body.**

Leucippus and Zeno say, that it is a body and a fragment of the soul. Pythagoras, Plato, and Aristotle, that the spermatic faculty is incorporeal, as the mind is which moves the body; but the effused matter is corporeal. Strato and Democritus, that the essential power is a body; for it is like spirit.

## **Chapter V.**

### **Whether Women do Give a Spermatic Emission As Men Do.**

Pythagoras, Epicurus, and Democritus say, that women have a seminal projection, but their spermatic vessels are inverted; and it is this that makes them have a venereal appetite. Aristotle and Plato, that they emit a material moisture, as sweat we see produced by exercise and labor; but that moisture has no spermatic power. Hippo, that women have a seminal emission, but not after the mode of men; it contributes nothing to generation, for it falls outside of the matrix; and therefore some women without coition, especially widows, give the seed. They also assert that from men the bones, from women the flesh proceed.

## **Chapter VI.**

### **How it is that Conceptions are Made.**

Aristotle says, that conception takes place when the womb is drawn down by the natural purgation, and the monthly terms attract from the whole mass part of the purest blood, and this is met by the seed of man. On the contrary, there is a failure by the impurity and inflation of the womb, by fear and grief, by the weakness of women, or the decline of strength in men.

## **Chapter VII.**

### **After what Manner Males and Females are Generated.**

Empedocles affirms, that heat and cold give the difference in the generation of males and females. Hence is it, as histories acquaint us, that the first men originated from the earth in the eastern and southern parts, and the first females in the northern parts. Parmenides is of opinion perfectly contrariant. He affirms that men first sprouted out of the northern earth, for their bodies are more dense; women out of the southern, for theirs are more rare and fine. Hippo, that the more compacted and strong sperm, and the more

fluid and weak, discriminate the sexes. Anaxagoras and Parmenides, that the seed of the man is naturally cast from his right side into the right side of the womb, or from the left side of the man into the left side of the womb; there is an alteration in this course of nature when females are generated. Cleophanes, whom Aristotle makes mention of, assigns the generation of men to the right testicle, of women to the left. Leucippus gives the reason of it to the alteration or diversity of parts, according to which the man hath a yard, the female the matrix; as to any other reason he is silent. Democritus, that the parts common to both sexes are engendered indifferently; but the peculiar parts by the one that is more powerful. Hippo, that if the spermatic faculty be more effectual, the male, if the nutritive aliment, the female is generated.

### **Chapter VIII.**

#### **By what Means it is that Monstrous Births are Effected.**

Empedocles believes that monsters receive their origination from the abundance or defect of seed, or from its division into parts which are superabundant, or from some disturbance in the motion, or else that there is an error by a lapse into an unsuitable receptacle; and thus he presumes he hath given all the causes of monstrous conceptions. Strato, that it comes through addition, subtraction, or transposition of the seed, or the distension or inflation of the matrix. And some physicians say that the matrix suffers distortion, being distended with wind.

### **Chapter IX.**

#### **How it Comes to Pass that a Woman's Too Frequent Conversation with a Man Hinders Conception.**

Diocles the physician says that either no genital sperm is projected, or, if there be, it is in a less quantity than nature requires, or there is no prolific faculty in it; or there is a deficiency of a due proportion of heat, cold, moisture, and dryness; or there is a resolution of the generative parts. The Stoics attribute sterility to the obliquity of the yard, by which means it is not able to ejaculate in a due manner, or to the unproportionable magnitude of the parts, the matrix being so contracted as not to have a capacity to receive. Erasistratus assigns it to the womb's being more callous or more carneous, thinner or smaller, than nature does require.

### **Chapter X.**

#### **Whence it is that One Birth Gives Two or Three Children.**

Empedocles affirms, that the superabundance of sperm and the division of it causes the bringing forth of two or three infants. Asclepiades, that it is performed from the excellent quality of the sperm, after the manner that from the root of one barleycorn two or three stalks do grow; sperm that is of this quality is the most prolific. Erasistratus, that superfetation may happen to women as to irrational creatures; for, if the womb be well purged and very clean, then there can be divers births. The Stoics, that it ariseth from the various receptacles that are in the womb: when the seed illapses into the first and second of them at once, then there are conceptions upon conception; and so two or three infants are born.

## **Chapter XI.**

### **Whence it is that Children Represent Their Parents and Progenitors.**

Empedocles says, that the similitude of children to their parents proceeds from the vigorous prevalency of the generating sperm; the dissimilitude from the evaporation of the natural heat it contains. Parmenides, that when the sperm falls on the right side of the womb, then the infant gives the resemblance of the father; if from the left, it is stamped with the similitude of the mother. The Stoics, that the whole body and soul give the sperm; and hence arise the likenesses in the characters and faces of the children, as a painter in his copy imitates the colors in a picture before him. Women have a concurrent emission of seed; if the feminine seed have the predominancy, the child resembles the mother; if the masculine, the father.

## **Chapter XII.**

### **How it Comes to Pass that Children Have a Greater Similitude with Strangers Than with Their Parents.**

The greatest part of physicians affirm, that this happens casually and fortuitously; for, when the sperm of the man and woman is too much refrigerated, then children carry a dissimilitude to their parents. Empedocles, that a woman's imagination in conception impresses a shape upon the infant; for women have been enamoured with images and statues, and the children which were born of them gave their similitudes. The Stoics, that the resemblances flow from the sympathy and consent of minds, through the insertion of effluvias and rays, not of images or pictures.

## **Chapter XIII.**

### **Whence Ariseth Barrenness in Women, and Impotency in Men?**

The physicians maintain, that sterility in women can arise from the womb; for if it be after any ways thus affected, there will be a barrenness, — if it be more condensed, or more thin, or more hardened, or more callous, or more carneous; or it may be from languor, or from an atrophy or vicious condition of body; or, lastly, it may arise from a twisted or distorted position. Diocles holds that the sterility in men ariseth from some of these causes, — either that they cannot at all ejaculate any sperm, or if they do, it is less than nature doth require, or else there is no generative faculty in the sperm, or the genital members are flagging; or from the obliquity of the yard. The Stoics attribute the cause of sterility to the contrariant qualities and dispositions of those who lie with one another; but if it chance that these persons are separated, and there happen a conjunction of those who are of a suitable temperament, then there is a commixture according to nature, and by this means an infant is formed.

## **Chapter XIV.**

### **How it Arises that Mules are Barren.**

Alcmaeon says, that the barrenness of the male mules ariseth from the thinness of the genital sperm, that is, the seed is too chill; the female mules are barren, because the womb does not open its mouth (as he expresses it). Empedocles, the matrix of the mule is so small, so depressed, so narrow, so invertedly growing to the belly, that the sperm cannot be regularly ejaculated into it, and if it could, there would be no capacity to receive it. Diocles concurs in this opinion with him; for, saith he, in our anatomical dissection of mules we have seen that their matrices are of such configurations; and it is possible that there may be the same reason why some women are barren.

## **Chapter XV.**

### **Whether the Infant in the Mother's Womb be an Animal.**

Plato says, that the embryo is an animal; for, being contained in the mother's womb, motion and aliment are imparted to it. The Stoics say that it is not an animal, but to be accounted part of the mother's belly; like as we see the fruit of trees is esteemed part of the trees, until it be full ripe; then it falls and ceaseth to belong to the tree; thus it is with the embryo. Empedocles, that the embryo is not an animal, yet whilst it remains in the belly it breathes. The first breath that it draws as an animal is when the infant is newly born; then the child having its moisture separated, the extraneous air making an entrance into the empty places, a respiration is caused in the infant by the empty vessels receiving of it. Diogenes, that infants are nurtured in the matrix inanimate, yet they have a natural heat; but presently, when the infant is cast into the open air, its heat brings air into the lungs, and so it becomes an animal. Herophilus acknowledgeth that a natural,

but not an animal motion, and that the nerves are the cause of that motion; that then they become animals, when being first born they suck in something of the air.

## **Chapter XVI.**

### **How Embryos are Nourished, or How the Infant in the Belly Receives its Aliment.**

Democritus and Epicurus say, that the embryos in the womb receive their aliment by the mouth, for we perceive, as soon as ever the infant is born, it applies its mouth to the breast; in the wombs of women (our understanding concludes) there are little dugs, and the embryos have small mouths by which they receive their nutriment. The Stoics, that by the secundines and navel they partake of aliment, and therefore the midwife instantly after their birth ties the navel, and opens the infant's mouth, that it may receive another sort of aliment. Alcmaeon, that they receive their nourishment from every part of the body; as a sponge sucks in water.

## **Chapter XVII.**

### **What Part of the Body is First Formed in the Womb.**

The Stoics believe that the whole is formed at the same time. Aristotle, as the keel of a ship is first made, so the first part that is formed is the loins. Alcmaeon, the head, for that is the commanding and the principal part of the body. The physicians, the heart, in which are the veins and arteries. Some think the great toe is first formed; others affirm the navel.

## **Chapter XVIII**

### **Whence is it that Infants Born in the Seventh Month are Born Alive.**

Empedocles says, that when the human race took first its original from the earth, the sun was so slow in its motion that then one day in its length was equal to ten months, as now they are; in process of time one day became as long as seven months are; and there is the reason that those infants which are born at the end of seven months or ten months are born alive, the course of nature so disposing that the infant shall be brought to maturity in one day after that night in which it is begotten. Timaeus says, that we count not ten months but nine, by reason that we reckon the first conception from the stoppage of the menstruas; and so it may generally pass for seven months when really there are not seven; for it sometimes occurs that even after conception a woman is purged to some extent. Polybus, Diocles, and the Empirics, acknowledge that the eighth month gives a vital birth to the infant, though the life of it is more faint and languid; many therefore we see born in that month die out of mere weakness. Though we see many born in that

month arrive at the state of man, yet (they affirm) if children be born in that month, none wish to rear them.

Aristotle and Hippocrates, that if the womb is full in seven months, then the child falls from the mother and is born alive, but if it falls from her but is not nourished, the navel being weak on account of the weight of the infant, then it doth not thrive; but if the infant continues nine months in the womb, and then comes forth from the woman, it is entire and perfect. Polybus, that a hundred and eighty-two days and a half suffice for the bringing forth of a living child; that is, six months, in which space of time the sun moves from one tropic to the other; and this is called seven months, for the days which are over plus in the sixth are accounted to give the seventh month. Those children which are born in the eighth month cannot live, for, the infant then falling from the womb, the navel, which is the cause of nourishment, is thereby too much wrenched; and is the reason that the infant languishes and hath an atrophy. The astrologers, that eight months are enemies to every birth, seven are friends and kind to it. The signs of the zodiac are then enemies, when they fall upon those stars which are lords of houses; whatever infant is then born will have a life short and unfortunate. Those signs of the zodiac which are malevolent and injurious to generation are those pairs of which the final is reckoned the eighth from the first, as the first and the eighth, the second and the ninth, etc; so is the Ram unsociable with Scorpio, the Bull with Sagittarius, the Twins with the Goat, the Crab with Aquarius, the Lion with Pisces, the Virgin with the Ram. Upon this reason those infants that are born in the seventh or tenth months are like to live, but those in the eighth month will die.

## **Chapter XIX.**

### **Of the Generation of Animals, How Animals are Begotten, and Whether They are Obnoxious to Corruption.**

Those philosophers who entertain the opinion that the world had an original do likewise assert that all animals are generated and corruptible. The followers of Epicurus, who gives an eternity to the world, affirm the generation of animals ariseth from the various permutation of parts mutually among themselves, for they are parts of this world. With them Anaxagoras and Euripides concur:

Nothing dies,  
Different changes give their various forms.

Anaximander's opinion is, that the first animals were generated in moisture, and were enclosed in bark on which thorns grew; but in process of time they came upon dry land, and this thorny bark with which they were covered being broken, they lived only for a short space of time. Empedocles says, that the first generation of animals and

plants was by no means completed, for the parts were disjoined and would not admit of a union; the second preparation and for their being generated was when their parts were united and appeared in the form of images; the third preparation for generation was when their parts mutually amongst themselves gave a being to one another; the fourth, when there was no longer a mixture of like elements (as earth and water), but a union of animals among themselves, — in some the nourishment being made dense, in others female beauty provoking a desire of spermatic motion. All sorts of animals are discriminated by their proper temperament and constitution; some are carried by a proper appetite and inclination to water, some, which partake of a more fiery quality, to live in the air those that are heavier incline to the earth; but those animals whose parts are of a just temperament are fitted equally for all places.

## **Chapter XX.**

### **How Many Species of Animals There are, and Whether All Animals Have the Endowments of Sense and Reason.**

There is a certain treatise of Aristotle, in which animals are distributed into four kinds, terrestrial, aqueous, fowl, and heavenly; and he calls the stars and the world too animals, yea, and God himself he posits to be an animal gifted with reason and immortal. Democritus and Epicurus consider all animals rational which have their residence in the heavens. Anaxagoras says that animals have only that reason which is operative, but not that which is passive, which is justly styled the interpreter of the mind, and is like the mind itself. Pythagoras and Plato, that the souls of all those who are styled brutes are rational; but by the evil constitution of their bodies, and because they have a want of a discursive faculty, they do not conduct themselves rationally. This is manifested in apes and dogs, which have inarticulate voice but not speech. Diogenes, that this sort of animals are partakers of intelligence and air, but by reason of the density in some parts of them, and by the superfluity of moisture in others, they neither enjoy understanding nor sense; but they are affected as madmen are, the commanding rational part being defectuous and injured.

## **Chapter XXI.**

### **What Time is Required to Shape the Parts of Animals in the Womb.**

Empedocles believes, that the joints of men begin to be formed from the thirty-sixth day, and their shape is completed in the nine and fortieth. Asclepiades, that male embryos, by reason of a greater natural heat, have their joints begun to be formed in the twenty-sixth day, — many even sooner, — and that they are completed in all their parts on the fiftieth day; the parts of the females are articulated in two months, but by the defect of



heat are not consummated till the fourth; but the members of brutes are completed at various times, according to the commixture of the elements of which they consist.

## **Chapter XXII.**

### **Of what Elements Each of the Members of Us Men is Composed.**

Empedocles says, that the fleshy parts of us are constituted by the contemperation of the four elements in us; earth and fire mixed with a double proportion of water make nerves; but when it happens that the nerves are refrigerated where they come in contact with the air, then the nails are made; the bones are produced by two parts of water and the same of air, with four parts of fire and the same of earth, mixed together; sweat and tears flow from liquefaction of bodies.

## **Chapter XXIII.**

### **What are the Causes of Sleep and Death?**

Alcmaeon says, that sleep is caused when the blood retreats to the concourse of the veins, but when the blood diffuses itself then we awake and when there is a total retirement of the blood, then men die. Empedocles, that a moderate cooling of the blood causeth sleep, but a total remotion of heat from blood causeth death. Diogenes, that when all the blood is so diffused as that it fills all the veins, and forces the air contained in them to the back and to the belly that is below it, the breast being thereby more heated, thence sleep arises, but if everything that is airy in the breast forsakes the veins, then death succeeds. Plato and the Stoics, that sleep ariseth from the relaxation of the sensitive spirit, it not receiving such total relaxing as if it fell to the earth, but so that that spirit is carried about the intestine, parts of the eyebrows, in which the principal part has its residence; but when there is a total relaxing of the sensitive spirit, death ensues.

## **Chapter XXIV.**

### **When and from Whence the Perfection of a Man Commences.**

Heraclitus and the Stoics say, that men begin their completeness when the second septenary of years begins, about which time the seminal serum is emitted. Trees first begin their perfection when they give their seeds; till then they are immature, imperfect, and unfruitful. After the same manner a man is completed in the second septenary of years, and is capable of learning what is good and evil, and of discipline therein.

## **Chapter XXV.**

## **Whether Sleep or Death Appertains to the Soul or Body.**

Aristotle's opinion is, that both the soul and body sleep; and this proceeds from the evaporation in the breast, which doth steam and arise into the head, and from the aliment in the stomach, whose proper heat is cooled in the heart. Death is the perfect refrigeration of all heat in body; but death is only of the body, and not of the soul, for the soul is immortal. Anaxagoras thinks, that sleep makes the operations of the body to cease; it is a corporeal passion and affects not the soul. Death is the separation of the soul from the body. Leucippus, that sleep is only of the body; but when the smaller particles cause excessive evaporation from the soul's heat, this makes death; but these affections of death and sleep are of the body, not of the soul. Empedocles, that death is nothing else but separation of those fiery parts by which man is composed, and according to this sentiment both body and soul die; but sleep is only a smaller separation of the fiery qualities.

## **Chapter XXVI.**

### **How Plants Increase.**

Plato and Empedocles believe, that plants are animals, and are informed with a soul; of this there are clear arguments, for they have tossing and shaking, and their branches are extended; when the woodmen bend them they yield, but they return to their former straightness and strength again when they are let loose, and even carry up weights that are laid upon them. Aristotle doth grant that they live, but not that they are animals; for animals are affected with appetite, sense, and reason. The Stoics and Epicureans deny that they are informed with a soul; by reason that all sorts of animals have either sense, appetite, or reason; but plants act fortuitously, and not by means of any soul. Empedocles, that the first of all animals were trees, and they sprang from the earth before the sun in its motion enriched the world, and before day and night were distinguished; but by the harmony which is in their constitution they partake of a masculine and feminine nature; and they increase by that heat which is exalted out of the earth, so that they are parts belonging to it, as embryos in the womb are parts of the womb. Fruits in plants are excrescences proceeding from water and fire; but the plants which lack water, when this is dried up by the heat of summer, shed their leaves; whereas they that have plenty thereof keep their leaves on, as the olive, laurel, and palm. The differences of their moisture and juice arise from the difference of particles and various other causes, and they are discriminated by the various particles that feed them. And this is apparent in vines for the excellence of wine flows not from the difference in the vines, but from the soil from whence they receive their nutriment.

## **Chapter XXVII**

## **Of Nutrition and Growth.**

Empedocles believes, that animals are nourished by the remaining in them of that which is proper to their own nature; they are augmented by the application of heat; and the subtraction of either of these makes them to languish and decay. The stature of men in this present age, if compared with the magnitude of those men which were first produced, is only a mere infancy.

### **Chapter XXVIII.**

#### **Whence it is that in Animals There are Appetites and Pleasures.**

Empedocles says that the want of those elements which compose animals gives to them appetite, and pleasures spring from humidity. As to the motions of dangers and such like things as perturbations, etc. . . .

### **Chapter XXIX**

#### **What is the Cause of a Fever, or Whether it is an Affection of the Body Annexed to a Primary Passion**

Erasistratus gives this definition of a fever: A fever is a quick motion of blood, not produced by our consent, which enters into the vessels, the seat of the vital spirits. This we see in the sea; it is in a serene calm when nothing disturbs it, but is in motion when a violent preternatural wind blows upon it, and then it rageth and is circled with waves. After this manner it is in the body of man; when the blood is in a nimble agitation, then it falls upon those vessels in which the spirits are, and there being in an extraordinary heat, it fires the whole body. The opinion that a fever is an appendix to a preceding affection pleaseth him. Diocles proceeds after this manner: Those things which are internal and latent are manifested by those which externally break forth and appear; and it is clear to us that a fever is annexed to certain outward affections, for example, to wounds, inflaming tumors, inguinary abscesses.

### **Chapter XXX.**

#### **Of Health, Sickness, and Old Age.**

Alcmaeon says that the preserver of health is an equal proportion of the qualities of heat, moisture, cold, dryness, bitterness, sweetness, and the other qualities; on the contrary, the prevailing empire of one above the rest is the cause of diseases and author of destruction. The direct cause of disease is the excess of heat or cold, the formal cause is excess or defect, the place is the blood or brain. But health is the harmonious

commixture of the elements. Diocles, that sickness for the most part proceeds from the irregular disposition of the elements in the body, for that makes an ill habit or constitution of it. Erasistratus, that sickness is caused by the excess of nourishment, indigestion, and corruptions; on the contrary, health is the moderation of the diet, and the taking that which is convenient and sufficient for us. It is the unanimous opinion of the Stoics that the want of heat brings old age, for (they say) those persons in whom heat more abounds live the longer. Asclepiades, that the Ethiopians soon grow old, and at thirty years of age are ancient men, their bodies being excessively heated and scorched by the sun; in Britain persons live a hundred and twenty years, on account of the coldness of the country, and because the people keep the fiery element within their bodies; the bodies of the Ethiopians are more fine and thin, because they are relaxed by the sun's heat, while they who live in northern countries are condensed and robust, and by consequence are more long lived.

## **Causes of natural phenomena**



### **I. What is the Reason that Sea-water nourishes not Trees?**

Is it not for the same reason that it nourishes not earthly animals? For Plato, Anaxagoras, and Democritus think plants are earthly animals. Nor, though sea-water be aliment to marine plants, as it is to fishes, will it therefore nourish earthly plants, since it can neither penetrate the roots, because of its grossness, nor ascend, by reason of its weight; for this, among many other things, shows sea-water to be heavy and terrene, because it more easily bears up ships and swimmers. Or is it because drought is a great enemy to trees? For sea-water is of a drying faculty; upon which account salt resists putrefaction, and the bodies of such as wash in the sea are presently dry and rough. Or is it because oil is destructive to earthly plants, and kills things anointed with it? But sea-water participates of much fatness; for it burns together with it. Wherefore, when men would quench fire, we forbid them to throw on sea-water. Or is it because sea-water is not fit to drink and bitter (as Aristotle says) through a mixture of burnt earth? For a lye is made by the falling of ashes into sweet water, and the dissolution ejects and corrupts what was good and potable, as in us men fevers convert the humors into bile. As for what woods and plants men talk of growing in the Red Sea, they bear no fruit, but are nourished by rivers casting up much mud; therefore they grow not at any great distance from land, but very near to it.

### **II. Why do Trees and Seeds thrive better with Rain than with Watering?**

Whether is it because (as Laitus thinks) showers, parting the earth by the violence of their fall, make passages, whereby the water may more easily penetrate to the root? Or cannot this be true; and did Laitus never consider that marsh-plants (as cat's-tail, pond-weeds, and rushes) neither thrive nor sprout when the rains fall not in their season; but it is true, as Aristotle said, rain-water is new and fresh, that of lakes old and stale? And what if this be rather probable than true? For the waters of fountains and rivers are ever fresh, new always arriving; therefore Heraclitus said well, that no man could go twice into the same river. And yet these very waters nourish worse than rain-water. But water from the heavens is light and aerial, and, being mixed with spirit, is the quicker passed and elevated into the plant, by reason of its tenuity. And for this very reason it makes bubbles when mixed with the air. Or does that nourish most which is soonest altered and overcome by the thing nourished? — for this very thing is concoction. On the contrary, inconcoction is when the aliment is too strong to be affected by the thing nourished. Now thin, simple, and insipid things are the most easily altered, of which number is rain-water, which is bred in the air and wind, and falls pure and sincere. But fountain-water, being assimilated to the earth and places through which it passes, is filled with many qualities which render it less nutritive and slower in alteration to the thing nourished. Moreover, that rain-water is easily alterable is an argument; because it sooner putrefies than either spring or river-water. For concoction seems to be putrefaction, as Empedocles says, —

- When in vine wood the water putrefies,
- It turns to wine, while under bark it lies.

Or, which may most readily be assigned for a reason, is it because rain is sweet and mild, when it is presently sent by the wind? For this reason cattle drink it most greedily, and frogs in expectation of it raise their voice, as if they were calling for rain to sweeten the marsh and to be sauce to the water in the pools. For Aratus makes this a sign of approaching rain,

- When father frogs, to watery snakes sweet food,
- Do croak and sing in mud, a wretched brood.

### **III. Why do Herdsmen set Salt before Cattle?**

Whether (as many think) to nourish them the more, and fatten them the better? For salt by its acrimony sharpens the appetite, and by opening the passages brings meat more easily to digestion. Therefore Apollonius, Herophilus's scholar, would not have lean persons, and such as did not thrive, be fed with sweet things and gruel, but ordered them to use pickles and salt things for their food, whose tenuity, serving instead of frication or sifting, might apply the aliment through the passages of the body. Or is it for health's sake that men give sheep salt to lick, to cut off the redundance of nutriment? For when they are over fat, they grow sick; but salt wastes and melts the fat. And this they observe so well, that they can more easily flay them; for the fat, which agglutinates and fastens

the skin, is made thin and weak by the acrimony. The blood also of things that lick salt is attenuated; nor do things within the body stick together when salts are mixed with them. Moreover, consider this, whether the cattle grow more fruitful and more inclined to coition; for bitches do sooner conceive when they are fed with salt victuals, and ships which carry salt are more pestered with mice, by reason of their frequent coition.

#### **IV. Why is the Water of Showers which falls in Thunder and Lightning fitter to Water Seeds? And they are therefore called Thunder-showers.**

Is it because they contain much spirit, by reason of their confusion and mixture with the air? And the spirit moving the humor sends it more upwards. Or is it because heat fighting against cold causes thunder and lightning? Whence it is that it thunders very little in winter, but in spring and autumn very much, because of the inequality of temper; and the heat, concocting the humor, renders it friendly and commodious for plants. Or does it thunder and lighten most in the spring for the aforesaid cause, and do the seeds have greater occasion for the vernal rains before summer? Therefore that country which is best watered with rain in spring, as Sicily is, produces abundance of good fruit.

#### **V. How comes it to pass, that since there be Eight Kinds of Tastes, we find the Salt in no Fruit whatever?**

Indeed, at first the olive is bitter, and the grape acid; one whereof afterward turns fat, and the other vinous. But the harshness in dates and the austere in pomegranates turn sweet. Some pomegranates and apples have only a simple acid taste. The pungent taste is frequent in roots and seeds.

Is it because a salt taste is never natural, but arises when the rest are corrupt? Therefore such plants and seeds as are nourished receive no nourishment from salt; it serves indeed some instead of sauce, by preventing a surfeit of other nourishment. Or, as men take away saltiness and bitingness from the sea-water by distilling, is saltiness so abolished in hot things by heat? Or indeed does the taste (as Plato says) arise from water percolated through a plant, and does even sea-water percolated lose its saltiness, being terrene and of gross parts? Therefore people that dig near the sea happen upon wells fit to drink. Several also that draw the sea-water into waxen buckets receive it sweet and potable, the salt and earthy matter being strained out. And straining through clay renders sea-water potable, since the clay retains the earthy parts and does not let them pass through. And since things are so, it is very probable either that plants receive no saltiness extrinsically, or, if they do, they put it not forth into fruit; for things terrene and consisting of gross parts cannot pass, by reason of the straitness of the passages. Or may saltiness be reckoned a sort of bitterness? For so Homer says:

- Out of his mouth the bitter brine did flow,
- And down his body from his head did go.

Plato also says that both these tastes have an abstersive and colliquative faculty; but the salt does it less, nor is it rough. And the bitter seems to differ from the salt in abundance of heat, since the salt has also a drying quality.

#### **VI. What is the Reason that, if a Man frequently pass along Dewy Trees, those Limbs that touch the Wood are seized with a Leprosy?**

Whether (as Laitus said) that by the tenuity of the dew the moisture of the skin is fretted away? Or, as smut and mildew fall upon moistened seeds, so, when the green and tender parts on the superficies are fretted and dissolved by the dew, is a certain noxious taint carried and imparted to the most bloodless parts of the body, as the legs and feet, which there eats and frets the superficies? For that by Nature there is a corrosive faculty in dew sufficiently appears, in that it makes fat people lean; and gross women gather it, either with wool or on their clothes, to take down their flesh.

#### **VII. Why in Winter do Ships sail slower in Rivers, but do not so in the Sea?**

Whether, because the river-air, which is at all times heavy and slow, being in winter more condensed by the cold, does more resist sailing? Or is it long of the water rather than the air? For the piercing cold makes the water heavy and thick, as one may perceive in a waterclock; for the water passes more slowly in winter than in summer. Theophrastus talks of a well about Pangaeum in Thrace, how that a vessel filled with the water of it weighs twice as much in winter as it does in summer. Besides, hence it is apparent that the grossness of the water makes ships sail slower, because in winter river-vessels carry greater burthens. For the water, being made more dense and heavy, makes the more renitency; but the heat hinders the sea from being condensed or frozen.

#### **VIII. Why, since all other Liquors upon moving and stirring about grow cold, does the Sea by being tossed in Waves grow hot?**

Whether that motion expels and dissipates the heat of other liquors as a thing adscititious, and the winds do rather excite and increase the innate heat of the sea? Its transparentness is an argument of heat; and so is its not being frozen, though it is terrene and heavy.

#### **IX. Why in Winter is the Sea least salt and bitter to the Taste? For they say that Dionysius the Hydragogue reported this.**

Is it that the bitterness of the sea is not devoid of all sweetness, as receiving so many rivers into it; but, since the sun exhales the sweet and potable water thereof, arising to the top by reason of its levity, and since this is done in summer more than in winter, when it affects the sea more weakly by reason of the debility of its heat, that so in

winter a great deal of sweetness is left, which tempers and mitigates its excessive poisonous bitterness? And the same thing befalls potable waters; for in summer they are worse, the sun wasting the lightest and sweetest part of them. And a fresh sweetness returns in winter, of which the sea must needs participate, since it moves, and is carried with the rivers into the sea.

**X. Why do Men pour Sea-water upon Wine, and say the Fishermen had an Oracle given them, whereby they were bid to dip Bacchus into the Sea? And why do they that live far from the Sea cast in some Zacynthian Earth toasted?**

Whether that heat is good against cold? Or that it quenches heat, by diluting the wine and destroying its strength? Or that the aqueous and aerial part of wine (which is therefore prone to mutation) is stayed by the throwing in of terrene parts, whose nature it is to constipate and condense? Moreover, salts with the sea-water, attenuating and colliquating whatever is foreign and superfluous, suffer no fetidness or putrefaction to breed. Besides, the gross and terrene parts, being entangled with the heavy and sinking together, make a sediment or lees, and so make the wine fine.

**XI. Why are they Sicker that Sail on the Sea than they that Sail in fresh Rivers, even in Calm Weather?**

Of all the senses, smelling causes nauseousness the most, and of all the passions of the mind, fear. For men tremble and shake and bewray themselves upon apprehension of great danger. They that sail in a river are troubled with neither of these. And the smell of sweet and potable water is familiar to all, and the voyage is without danger. On the sea an unusual smell is troublesome; and men are afraid, not knowing what the issue may be. Therefore tranquillity abroad avails not, while an estuating and disturbed mind disorders the body.

**XII. Why does pouring Oil on the Sea make it Clear and Calm?**

Is it for that the winds, slipping the smooth oil, have no force, nor cause any waves? This may be probably said in respect of things external; but they say that divers take oil in their mouths, and when they spout it out they have light at the bottom, and it makes the water transparent; so that the slipping of the winds will not hold good here for an argument. Therefore it is to be considered, whether the sea, which is terrene and uneven, is not compacted and made smooth by the dense oil; and so the sea, being compact in itself, leaves passes, and a pellucidity penetrable by the sight. Or whether that the air, which is naturally mixed with the sea, is lucid, but by being troubled grows unequal and shady; and so by the oil's density, smoothing its inequality, the sea recovers its evenness and pellucidity.



### **XIII. Why do Fishermen's Nets rot more in Winter than in Summer, since other things rot more in Summer?**

Is not that the cause which Theophrastus assigns, — that heat (to wit) shuns the cold, and is constrained by it on every side? Hence the waters are hottest in the bottom of the sea. And so it is on land; for springs are hotter in winter, and then lakes and rivers send up most vapors, because the heat is compelled to the bottom by the prevailing cold. Or it may be, nets do not rot at that time more than at another; but being frozen and dried in the cold, since they are therefore the more easily broken by the waves, they are liable to something like putrefaction and rottenness. And they suffer most in the cold (as strained cords are aptest to break in such a season), because then there be most frequent storms at sea. Therefore fishermen guard their nets with certain tinctures, for fear they should break. Otherwise a net, neither tinged nor daubed with any thing, might more easily deceive the fish; since line is of an air color, and is not easily discerned in the sea.

### **XIV. Why do the Dorians pray for bad making of their hay?**

Is it because hay rained upon is never well made? For the grass is cut down green and not dry, wherefore it putrefies when wet with rain water. But when before harvest it rains upon corn, this is a help to it against the hot south winds; which otherwise would not let the grain fill in the ear, but by their heat would hinder and destroy all coalition, unless by watering the earth there came a moisture to cool and moisten the ear.

### **XV. Why is a fat and deep Soil fruitful of Wheat, and a lean Soil of Barley?**

Is it because a stronger grain needs more nourishment, and a weaker a light and thin one? Now barley is weaker and laxer than wheat, therefore it affords but little nourishment. And, as a farther testimony to this reason, wheat, that is ripe in three months, grows in dryer ground; because it is juiceless, and stands in need of less nourishment, and therefore is more easily brought to perfection.

### **XVI. Why do Men say, Sow wheat in Clay and Barley in Dust?**

Is the reason (as we said) because wheat takes up more nourishment; and barley cannot bear so much, but is choked with it? Or does wheat, because it is hard and ligneous, thrive better when it is softened and loosened in a moist soil; and barley at the first in a dry soil, because of its rarity? Or is the one temperament congruous and harmless to wheat, because it is hot; and the other to barley, because it is cold? Or are men afraid to sow wheat in a dry soil, because of the ants, which presently lie in wait for it; but they cannot so easily deal with barley nor carry it away, because it is a larger grain?

### **XVII. Why do Men use the Hair of Horses rather than of Mares for Fishing-Lines?**

Is it that the males are stronger in those parts, as well as in others, than the females? Or is it that the females spoil the hair of their tails by their staling?

### **XVIII. Why is the Sight of a Cuttle-fish a Sign of a great Storm?**

Is it because all fishes of the soft kind cannot endure cold, by reason of their nakedness and tenderness? For they are covered neither with shell, skin, or scale, though within they have hard and bony parts. Hence the Greeks call them soft fish. Therefore they easily perceive a storm coming, since they are so soon affected by the cold. When the polypus gets to shore and embraces the rocks, it is a sign the wind is rising; but the cuttle-fish jumps up, to shun the cold and the trouble of the bottom of the sea; for, of all soft fishes, she is the tenderest and soonest hurt.

### **XIX. Why does the Polypus change Color?**

Whether, as Theophrastus writes, because it is an animal by nature timorous; and therefore, being disturbed, it changes color with its spirit, as some men do, of whom it is said, an ill man ever changes color? But though this may serve as a reason for changing its color, it will not for the imitation of colors. For the polypus does so change its color, that it is of the color of every stone it comes nigh. Hence that of Pindar, Mind the color of the marine beast, and so converse cunningly in all cities; and that of Theognis:

- Put on a mind like th' polyp fish, —
- And learn so to dissemble, —
- Which of the rock whereto it sticks
- The color doth resemble.

And they say, that such as are excellent at craftiness and juggling have this in their eye, — that they may the better cheat them they have to deal withal, — ever to imitate the polypus. Some think the polypus can use her skin as a garment, and can put it on or off at pleasure. But if fear occasions this change in the polypus, is not something else more properly the cause? Let us consider what Empedocles says, that effluvia proceed from all things whatever. For not only animals, plants, the earth and sea, but stones, and even brass and iron, do continually send out many effluvia. For all things corrupt and smell, because there runneth always something from them, and they wear continually; insomuch that it is thought that by these effluvia come all attractions and insultations, some supposing embraces, others blows, some impulses, others circutions. But especially about the sea rocks, when they are wet and cool by the waves (as is most likely), constantly some small particles are washed off, which do not incorporate with other bodies, but either pass by the smaller passages, or pass through the larger. Now the flesh of the polypus, as one may judge by the eye, is hollow, full of pores, and capable of effluvia. When therefore she is afraid, as her spirit changes she changes

herself, and by straitening and contracting her body, she encloses the neighboring effluvia. And, as a good token of this argument, the polypus cannot imitate the color of every thing he comes near, nor the chameleon of any thing that is white; but each of these creatures is assimilated only to those things to whose effluvia it has pores proportionable.

**XX. What is the Reason, that the Tears of wild Boars are sweet, and the Tears of the Hart salt and hurtful?**

The reason seems to be the heat and cold of these animals. For the hart is cold, and the boar is very hot and fiery; therefore the one flies from, the other defends himself against, his pursuers. Now when great store of heat comes to the eyes (as Homer says, with horrid bristles, and eyes darting fire), tears are sweet. Some are of Empedocles's opinion, who thought that tears proceed from the disturbance of the blood, as whey does from the churning of milk; since therefore boar's blood is harsh and black, and hart's blood thin and watery, it is consentaneous that the tears, which the one sheds when excited to anger, and the other when dejected with fear, should be of the same nature.

**XXI. Why do tame Sows farrow often, some at one time and others at another; and the wild but once a Year, and all of them about the same time at the beginning of summer, whence it is said, —**

The wild sow farrowing, that night falls no rain?

Is it because of plentiful feeding, as in very truth fulness doth produce wantonness? For abundance of nourishment breeds abundance of seed both in animals and plants. Now wild sows live by their own toil, and that with fear; the tame have always food enough, either by nature or given them. Or may it not be ascribed to their rest and exercise? For the tame do rest and go not far from their keepers; the wild get to the mountains, and run about, by which means they waste the nutriment, and consume it upon the whole body. Therefore either through continual converse, or abundance of seed, or because the females feed in herds with the males, the tame sows call to mind coition and stir up lust, as Empedocles talks of men. But in wild sows, which feed apart, desire is cold and dull for want of love and conversation. Or is it true, what Aristotle says, that Homer called the wild boar *χλούνης*, because he had but one stone? For most boars spoil their stones (he says) by rubbing them against stumps of trees.

**XXII. Why are the Paws of Bears the sweetest and pleasantest Food?**

Because the flesh of those parts of the body which concoct aliment the best is sweetest; and that concocts best which transpires most by motion and exercise. But the bear uses the fore-feet most in going and running, and in managing of things, as it were with hands.

### **XXIII. Why are the Steps of wild Beasts most difficultly Traced in Spring-time?**

Whether the dogs, as Empedocles says, “with noses find the steps of all wild beasts,” and draw in those effluvia which the beasts leave in the ground; but the various smells of plants and flowers lying over the footsteps do in spring-time obscure and confound them, and put the dogs to a loss at winding them? Therefore about Etna in Sicily no man keeps any hunting dogs, because abundance of wild marjoram flourishes and grows there the year round, and the perpetual fragrancy of the place destroys the scent of the wild beasts. There is also a tale, how Proserpine, as she was gathering flowers thereabout, was ravished by Pluto; therefore people, revering that place as an asylum, do not catch any creature that feeds thereabout.

### **XXIV. Why are the Tracks of Wild Beasts worse Scented about the Full Moon?**

Whether for the foresaid cause? For the full moons bring down the dews; and therefore Alcman calls dew the daughter of Jove and Luna in a verse of his,

Fed by the dew, bred by the Moon and Jove.

For dew is a weak and languid rain, and there is but little heat in the moon; which draws water from the earth, as the sun does; but because it cannot raise it on high, it soon lets it fall.

### **XXV. Why does Frost make Hunting difficult?**

Whether is it because the wild beasts leave off going far abroad by reason of the cold, and so leave but few signs of themselves? Therefore some say, beasts spare the neighboring places, that they may not be sore put to it by going far abroad in winter, but may always have food ready at hand. Or is it because that for hunting the track alone is not sufficient, but there must be scent also? And things gently dissolved and loosened by heat afford a smell, but too violent cold binds up the scent, and will not let it reach the sense. Therefore they say that unguents and wine smell least in winter and cold weather; for the then concrete air keeps the scent in, and suffers it not to disperse.

### **XXVI. What is the Reason that Brutes, when they ail any thing, seek and pursue Remedies, and are often cured by the use of them?**

Dogs eat grass, to make them vomit bile. Swine seek craw-fish, because the eating of them cures the headache. The tortoise, when he has eaten a viper, feeds on wild marjoram. They say, when a bear has surfeited himself and his stomach grows nauseous, he licks up ants, and by devouring them is cured. These creatures know such things neither by experience nor by chance.

Whether, as wax draws the bee, and carcasses the vulture afar off by the scent, do craw-fish so draw swine, wild marjoram the tortoise, and ants the bear, by smells and

effluvia accommodated to their nature, they being prompted altogether by sense, without any assistance from reason? Or do not the temperaments of the body create appetites in animals, while diseases create these, producing divers acrimonies, sweetnesses, and other unusual and absurd qualities, the humors being altered; as is plain in women with child, who eat stones and earth? Therefore skilful physicians take their prognostic of recovery or death from the appetites of the sick. For Mnesitheus the physician says that, in the beginning of a disease of the lungs, he that craves onions recovers, and he that craves figs dies; because appetites follow the temperament, and the temperament follows diseases. It is therefore probable that beasts, if they fall not into mortal diseases, have such a disposition and temper, that by following their temper they light on their remedies.

### **XXVII. Why does Must, if the Vessel stand in the Cold, continue long sweet?**

Is it because the changing of the sweet must into wine is concoction, but cold hinders concoction, because this is caused by heat? Or, on the contrary, is the proper taste of the grape sweet, and is it then said to be ripe, when the sweetness is equally diffused all over it; but does cold, not suffering the heat of the grape to exhale, and keeping it in, conserve the sweetness of the grape? And this is the reason that, in a rainy vintage, must ferments but little; for fermentation proceeds from heat, which the cold does check.

### **XXVIII. Why, of all Wild Beasts, does not the Boar bite the Toil, although both Wolves and Foxes do this?**

Is it because his teeth stand so far within his head, that he cannot well come at the thread? For his lips, by reason of their thickness and largeness, meet close before. Or does he rather rely on his paws and mouth, and with those rend the toil, and with this defend himself against the hunters? His chief refuge is rolling and wallowing; therefore, rather than stand gnawing the toil, he rolls often about, and so clears himself, having no occasion for his teeth.

### **XXIX. What is the Reason that we admire Hot Waters (*i. e.* Baths) and not Cold; since it is plain that Cold is as much the cause of one sort as Heat is of the other?**

It is not (as some are of opinion) that heat is a quality, and cold only a privation of that quality, and so that an entity is even less a cause than a non-entity. But we do it because Nature has attributed admiration to what is rare, and she puts men upon enquiry how any thing comes to pass that seldom happens. As Euripides saith,

- Behold the boundless Heaven on high,
- Bearing the earth in his moist arms, —

what wonders he brings out by night, and what beauty he shows forth by day! . . . The rainbow and the varied beauty of the clouds by day, and the lights which burst forth by night . . .

**XXX. Why are Vines which are rank of leaves, but otherwise fruitless, said  
τραγᾶν?**

Is it because very fat goats (τράγοι) are less able to procreate, nay, scarce able to use coition, by reason of their fatness? Seed is the superfluity of the aliment which is allotted to the body: now, when either an animal or a plant is of a very strong constitution and grows fat, it is a sign that all the nourishment is spent within, and that there is little and base excrement, or none at all.

**XXXI. Why does the Vine irrigated with Wine die, especially the very Wine made  
from its own Grapes?**

Is it as baldness happens to great wine-bibbers, the heat of the wine evaporating the moisture? Or, as Empedocles saith, “the putrefied water in the wood becomes wine beneath the bark,” . . . thus, when the vine is outwardly irrigated with wine, it is as fire to the vine, and destroys the nutritive faculty. Or, because wine is obstructive, it gets into the roots, stops the passages, and so hinders any moisture from coming to the plant to make it grow and thrive. Or, it may seem contrary to Nature that that should return into the vine which came out of it; for whatsoever moisture comes from plants can neither nourish nor be again a part of the plant.

**XXXII.**

**Why doth the Palm alone of all trees bend Upward when a weight is laid  
thereupon?**

Is it that the fiery and spiritual power which it hath, being once provoked and (as it were) angered, putteth forth itself so much the more, and mounteth upward? Or is it because the weight, forcing the boughs suddenly, oppreseth and keepeth down the airy substance which they have, and driveth all of it inward; but the same afterwards, having resumed strength again, maketh head afresh, and more eagerly withstandeth the weight? Or, lastly, is it that the softer and more tender branches, not able to sustain the violence at first, so soon as the burden resteth quiet, by little and little lift up themselves, and make a show as if they rose up against it?

**XXXIII. What is the Reason that Pit-water is less nutritive than either that which  
ariseth out of Springs or that which falleth down from Heaven?**

Is it because it is more cold, and withal hath less air in it? Or because it containeth much salt from the earth mingled therewith? — now it is well known that salt above all other things causeth leanness. Or because standing still, and not exercised with running and stirring, it getteth a certain malignant quality, which is hurtful to both plants and animals, and is the cause that it is neither well concocted nor able to feed and nourish any thing? Hence it is that all dead waters of pools are unwholesome, for that they cannot digest and despatch those harmful qualities which they borrow of the evil property of the air or of the earth.

**XXXIV. Why is the West Wind held commonly to be the Swiftest, according to this Verse of Homer:**

- Let us likewise bestir our feet,
- As fast as Western winds do fleet.

Is it not because this wind is wont to blow when the sky is very well cleansed, and the air is exceeding clear and without all clouds? — for the thickness and impurity of the air doth not a little impeach and interrupt the course of the winds. Or is it rather because the sun, striking through a cold wind with his beams, is the cause that it passeth the faster away? — for whatsoever of cold is drawn in by the force of the winds, when the same is overcome by heat, as it were its enemy, we must think, is driven and set forward further and with greater celerity.

**XXXV. Why cannot Bees abide Smoke?**

Whether is it because the passages of their vital spirits are exceeding strait, and, if it chance that smoke be gotten into them and there kept in and intercepted, it is enough to stop the poor bees' breath, — yea, and to strangle them quite? Or is not the acrimony and bitterness (think you) of the smoke in cause? — for bees are delighted with sweet things, and in very truth they have no other nourishment; and therefore no marvel if they detest and abhor smoke, as a thing for the bitterness most adverse and contrary unto them. Therefore honey-masters, when they make a smoke for to drive away bees, are wont to burn bitter herbs, as hemlock, centaury, &c.

**XXXVI. Why will Bees sooner Sting those who newly before have committed Whoredom?**

Is it not because it is a creature that wonderfully delighteth in purity, cleanliness, and elegancy, and withal hath a marvellous quick sense of smelling? Because therefore such unclean dealings between man and woman are wont to leave behind much filthiness and impurity, the bees both sooner find them out and also conceive the greater hatred against them. Hereupon it is that in Theocritus the shepherd pleasantly sendeth Venus away unto Anchises to be well stung with bees for her adultery:

- Now to mount Ida, to Anchises go,
- Where mighty oaks and cypresses do grow;
- Where hives and trees with honey sweet abound,
- And both with humming noise of bees resound.

And Pindar saith: “Thou little creature, who honey-combs dost frame, and with thy sting hast pricked false impure Rhoecus for his lewd villanies.”

### **XXXVII. Why do Dogs follow after a Stone that is thrown at them and bite it, letting the Man alone who flung it?**

Is it because he can comprehend nothing by imagination nor call a thing to mind, which are gifts and virtues proper to man alone; and therefore, seeing he cannot discern the party that offered him injury, he supposeth that to be his enemy which seemeth in his eye to threaten him, and of it he goes about to be revenged? Or is it that he thinks the stone, while it runs along the ground, to be some wild beast, and according to his nature he intendeth to catch it first; but afterwards, when he seeth himself deceived and put besides his reckoning, he setteth upon the man? Or rather, doth he not hate the man and the stone both alike, but pursueth that only which is next unto him?

### **XXXVIII. Why at a certain time of the year do all She-wolves Whelp within the compass of twelve days?**

Antipater in his History of Animals affirms, that shewolves exclude forth their young ones about the time that mast trees shed their blossoms, for upon the taste thereof their wombs open; but if there be none of such blooms to be had, then their young die within the body and never come to light. Moreover, he saith, those countries which bring not forth oaks and mast are never troubled nor spoiled with wolves. Some attribute all this to a tale that goes of Latona; who being with child, and finding no abiding place of rest and safety by reason of Juno for the space of twelve days, went to Delos, and, being transmuted by Jupiter into a wolf, obtained at his hands that all wolves for ever after might within that time be delivered of their young.

### **XXXIX. How cometh it that Water, seeming White aloft, showeth to be Black in the bottom?**

Is it because depth is the mother of darkness, so that it doth dim and mar the sunbeams before they can descend so low as it? As for the uppermost superficies of the water, because it is immediately affected by the sun, it must needs receive the white brightness of the light; the which Empedocles verily approveth in these verses:

- A river in the bottom seems
- By shade of color black;
- The like is seen in caves and holes,



- By depth, where light they lack.

Or, since the bottom of the sea and of great rivers is often full of mud, doth it by reflection of the sunbeams represent the like color that the said mud hath? Or is it more probable that the water toward the bottom is not pure and sincere, but corrupted with an earthy quality, — as continually carrying with it somewhat of that by which it runneth and wherewith it is stirred, — and the same settling once to the bottom causeth it to be more troubled and less transparent?